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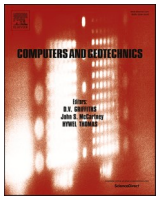

Research Paper

Stabilized unified multi-physics framework for methane hydrate production

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ABSTRACT

The safe and efficient extraction of methane hydrate reservoirs relies heavily on the accurate prediction of intricate interactions among thermal (T), hydraulic (H), mechanical (M), and chemical (C) responses. However, standard finite element simulations for methane hydrate typically lack a unified framework capable of accommodating the various coupling modes (e.g., TH, HM, THM, THMC) required across diverse engineering scenarios. Moreover, they often suffer from numerical instabilities, particularly in near-undrained conditions and convection-dominated flows. To address these challenges, this study proposes a unified multi-physics coupling framework designed to balance production enhancement strategies with reservoir stability. The main contributions of this framework include: (1) the architecture allows for dynamic spatiotemporal switching and coupling of different multi-physics schemes, tailoring the simulation to specific extraction phases and distinct reservoir strata. (2) an advanced hybrid stabilization strategy is incorporated to eradicate numerical artifacts. Specifically, the bubble-enhanced edge-based smoothed finite element method (bES-FEM) is employed to satisfy the inf-sup condition and eliminate pressure oscillations, while the Streamline Upwind Petrov-Galerkin (SUPG) technique is integrated to suppress instabilities in convection-dominated transport. (3) the explicit–implicit scheme is further implemented to efficiently handle the strong non-linearities inherent in hydrate dissociation. Following rigorous validation against benchmarks, the framework is then applied to field-scale methane extraction models. The results demonstrate the method's proficiency in capturing complex THMC phenomena, revealing the thermal field as a dominant controlling factor and elucidating the positive coupled effects of lower-strata depressurization and heating on reservoir behavior. Consequently, this work provides a reliable computational tool for investigating the complex multi-physics mechanisms underlying methane hydrate production.

1. Introduction

Methane hydrates, widely distributed across deep-marine sediments and permafrost regions, represent a massive and highly promising unconventional energy resource (Mu et al., 2026; Chen et al., 2025; Du et al., 2025; Gong et al., 2025). However, the safe and efficient extraction of methane hydrate-bearing sediments (MHBS) poses formidable theoretical and engineering challenges. These difficulties primarily stem from the intricate thermal (T), hydraulic (H), mechanical (M), and chemical (C) couplings inherent in the hydrate dissociation process. Specifically, the phase transition from solid hydrates to multiphase fluids triggers a cascade of complex interdependent responses: pore pressure fluctuations induce sediment deformation (HM coupling (Li et al., 2025; Hu et al., 2023), fluid flow redistributes the local temperature field (TH coupling (Zhang et al., 2025; Lin et al., 2025), and continuous dissociation inherently degrades the mechanical strength of the soil matrix (THMC coupling (Kimoto et al., 2010). Recent experimental studies have further confirmed that hydrate production by depressurization is strongly governed by thermodynamic evolution, ice formation, sediment heat transfer, dissociation kinetics, and water–gas production behavior in water-saturated deposits (Li et al., 2026; Song et al., 2022; Tian et al., 2022; Zheng et al., 2021). These observations

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provide important physical evidence for the necessity of a coupled THMC modeling framework. Consequently, developing a robust and unified multi-physics coupling framework is of great significance for accurately simulating and investigating the complex THMC interactions during hydrate production.

Over the past decades, various numerical solvers have been employed for simulating the multi-physics coupling processes of MHBS, such as the TOUGH series (Wan et al., 2026; Rutqvist et al., 2010; Moridis, 2003; Moridis et al., 2005; Kim et al., 2022), FLAC-based simulators (Kim et al., 2018), MH21-HYDRES (Kurihara, M, Ouchi, H, Y. Masuda, H. Narita, Y. Okada, Assessment of gas productivity of natural methane hydrates using MH21 Reservoir Simulator, in: Proc. AAPG Hedberg Conf. Vanc. BC Can., 2004), DEHydrate (Zhang et al., 2024), COMSOL Multi-physics solvers (Zhang and Lan, 2025; Liu et al., 2026; Al-Weheibi et al., 2017); and other codebases (Tang et al., 2026; Zhang et al., 2023; Zhang et al., 2024; Li et al., 2024; Xu et al., 2025; He et al., 2025; Liu et al., 2025; Zhang et al., 2025). These platforms have significantly advanced the understanding of hydrate dissociation, multiphase flow, heat transfer, and geomechanical responses. Nevertheless, the present work is not intended to serve as a general-purpose replacement for these established simulators. Instead, it targets a specific methodological need in methane hydrate production: a stabilized finite-element framework that can flexibly activate different coupling levels and numerical stabilization strategies within a single computational architecture. Most existing solvers are designed either for specific lower-order couplings (e.g., HC (Verigin et al., 1980; Zhang and Sheng, 2026); HMC (Wang et al., 2024; Xu et al., 2022) or THC (Yin et al., 2018a,b; Masuda, 1999) or exclusively for full THMC problems (Yu et al., 2026a,b). They often lack a unified framework capable of seamlessly transitioning between different coupling modes (from TM/TH/HM to fully coupled THMC (Yu et al., 2026a,b; Ding et al., 2026), preventing the flexible application of physical processes either to the entire model or to specific local sub-domains. This limitation is particularly acute in deep-sea methane hydrate exploitation, where significant spatial heterogeneity necessitates differentiated coupling strategies. For instance, the hydrate-bearing reservoir demands a rigorous fully coupled THMC analysis, the overlying and underlying saturated sediments may only require THM coupling, and the shallow seawater region can often be simplified to a TH problem. Furthermore, from an engineering perspective, efficiency and flexibility are paramount: preliminary assessments might only require simplified HM analysis, while detailed safety evaluations demand full THMC coupling. Therefore, a unified framework capable of switching among TH, HM, THM, and THMC formulations and assigning them to different subdomains is necessary for balancing physical fidelity and computational efficiency in heterogeneous hydrate exploitation scenarios.

Beyond the architectural limitations, standard finite element (FE) simulations of MHBS face severe numerical challenges, particularly when dealing with the strong non-linearities of phase transitions. Two specific issues undermine the reliability of conventional methods. First, MHBS might behave as nearly incompressible media, especially in undrained or low-permeability zones during rapid dissociation. In these scenarios, standard low-order finite elements fail to satisfy the Ladyzhenskaya-Babuška-Brezzi (inf-sup) condition (Brezzi and Bathe, 1990; White and Borja, 2008), leading to spurious pore pressure oscillations (volumetric locking). To mitigate this, numerous strategies have been developed, such as finite incremental calculus (Oñate, 2000; Preisig and Prévost, 2011), fractional step methods (Pastor et al., 2000), and polynomial pressure projections (White and Borja, 2008; Wei et al., 2016; Zhao and Choo, 2020). However, these methods typically involve either the introduction of artificial stabilization terms, complex element constructions, or the adoption of higher-order shape functions, which inevitably increase computational burdens. Second, the rapid release of gas and water creates convection-dominated flow regimes. The standard Galerkin formulations (Brooks and Hughes, 1982; Cao et al., 2025), which discretizes non-self-adjoint convective operators into skew-symmetric matrices, is prone to inducing non-physical spatial oscillations. Although stabilization methodologies like the Taylor-Galerkin (Donea, 1984; Hawken et al., 1990), Variational MultiScale (Ahmed et al., 2017; Ahmed and Rubino, 2018), or Artificial Diffusion methods (Firooz et al., 2026) have been introduced to ensure solution reliability, they often require complex parameter tuning or lack the versatility needed for coupled multi-physics. Such numerical instabilities are far from trivial, as they can severely distort the prediction of critical field variables, particularly pressure gradients and thermal fronts. This underscores the pressing need for a robust scheme capable of simultaneously resolving volumetric locking and convective oscillations without incurring prohibitive computational costs.

To address the dual challenges of architectural rigidity and numerical instability, this study proposes a stabilized unified multi-physics framework tailored for methane hydrate production. The innovation boundary of this work lies in the integration of three components within one finite-element framework: flexible activation of different coupling modes, hybrid bES-FEM/SUPG stabilization, and an explicit–implicit solution strategy for hydrate dissociation. The generalized architecture enables seamless transitions from simplified lower-order couplings to fully coupled THMC analyses and permits localized assignment of coupling schemes, such as THMC, THM, and TH, to different geological domains. This capability directly supports application scenarios in which heterogeneous strata, evolving production stages, and production-stability assessment must be handled in a consistent model. Beyond architectural flexibility, ensuring numerical stability under extreme physical conditions is paramount. To achieve this, we adopt the bubble-enhanced edge-based smoothed finite element method (bES-FEM) (Liu et al., 2009) in conjunction with the Streamline Upwind Petrov-Galerkin (SUPG) technique (Brooks and Hughes, 1982). While both bES-FEM and SUPG are established stabilization methods in general computational mechanics, their integration and specific application to the complex multi-physics of MHBS represent a key engineering contribution of this work. By enriching the ES-FEM displacement field with bubble functions (Nguyen-Xuan and Liu, 2013); the current formulation not only inherently satisfies the inf-sup condition to eliminate pressure oscillations in the incompressible limit, but also strikes an optimal balance by providing higher accuracy than lower-order elements and greater efficiency than higher-order elements. Concurrently, the integrated SUPG technique curbs the spatial instabilities inherent in convection-dominated problems. Furthermore, the explicit update of porosity and phase saturations, combined with the implicit solution of displacement, pore pressure, and temperature, reduces the size of the nonlinear coupled system while retaining robustness for strongly coupled THM responses. Consequently, the proposed framework provides a focused and competitive tool for methane hydrate production problems requiring flexible coupling, numerical stability, and computational efficiency within a unified FE implementation.

The remainder of the paper is organized to systematically present and apply the proposed framework. Section 2 establishes the unified mathematical formulation governing the complex THMC processes in hydrate-bearing sediments. Building upon this theoretical foundation, Section 3 details the numerical implementation strategies. This includes the integration of explicit schemes for updating state variables, such as porosity and saturation, with implicit solvers for primary fields, alongside the specific derivations of the bES-FEM and SUPG stabilization techniques. Section 4 subsequently validates the accuracy and robustness of the framework through benchmark tests spanning various physical scenarios. Then, the unified framework is applied to a field-scale extraction model to compare coupling schemes, evaluate production strategies, and analyze the coupled responses of gas recovery and sediment deformation across varying burial depths in Section 5. Finally, Section 6 summarizes the core findings, offering actionable insights to guide the safe and efficient production of marine methane hydrate.

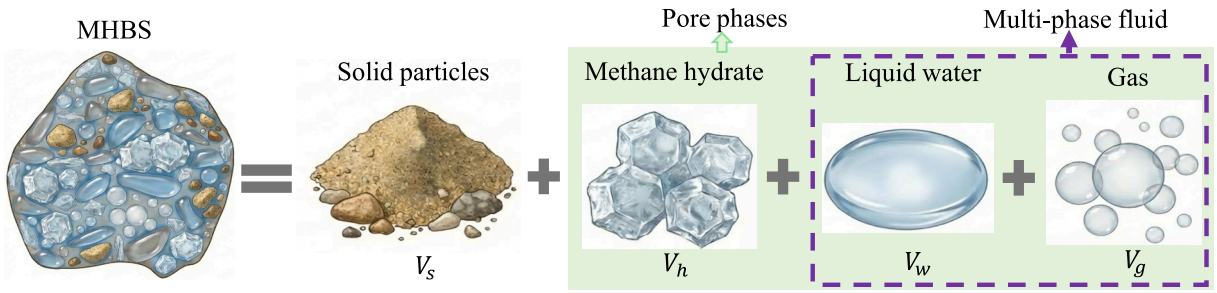

Fig. 1. MHBS medium composed of four distinct phases.

2. Mathematical formulation of THMC coupling

This section establishes a comprehensive mathematical framework for the fully coupled thermo-hydro-mechanical-chemical (THMC) processes. A key feature of the proposed model is its inherent flexibility, allowing for seamless switching among various coupling modes by selectively activating the corresponding governing equations. Within this framework, a hybrid explicit–implicit solution is employed in which phase saturations and porosity are updated explicitly while displacement, pore water pressure, and temperature (u-p-T) are solved implicitly. This approach enables seamless transitions between conventional THM coupling and the chemically coupled hydrate dissociation processes. The structure of this section is as follows: First, the basic assumptions and definitions for porous media are introduced. Next, the flow equations for water and gas are discussed based on Darcy's law. Subsequently, the strong and weak formulations for the conservation of mass, momentum, and energy are derived, along with their respective spatial and temporal discretization schemes.

2.1. Basic assumptions and definitions

MHBS is typically composed of multiple phases and components. To facilitate the characterization of the chemical field, the MHBS in this study is conceptualized as a four-phase system (see Fig. 1) comprising solid grains, water, gas, and methane hydrate. In the absence of methane hydrate dissociation, chemical coupling is disregarded, reducing the four-field THMC coupling formulation to a three-field THM coupling formulation (Gong et al., 2024; Banh and Lee, 2025). By further neglecting the specific physical fields (e.g., Thermal, Hydraulic or Mechanical field), simplified two-field coupling models can be derived, such as the HM, TM, or TH coupling models. Consequently, this THMC coupling model offers broad applicability and versatility for analyzing porous media. The following assumptions are adopted to construct this unified multi-physics framework:

1) The solid skeleton is deformable, with hydrate and water assumed to be incompressible (Yang et al., 2026) and the gas compressible.
2) The formation of ice and water vapor within the porous medium is neglected.
3) Relative motion between methane hydrate and solid grains is assumed to be negligible, and local thermal equilibrium is assumed among all phases.
4) The physical properties of all phases are isotropic, and the model is formulated under the infinitesimal strain assumption.

Based on the above assumptions, the multiphase porous medium is described using a representative elementary volume (REV) with a total volume V given by:

$$V = \sum V_\alpha \quad (\alpha = s, h, w, g)\,, \tag{1}$$

where s, h, w and g denote the phases of solid, hydrate, water, and gas, respectively. The pore space is occupied by the phases β in methane hydrate (h), water (w) and gas (g), while the porosity ϕ is defined as the ratio of the pore volume to the total volume (i.e., $\phi = \sum V_\beta / V$). Correspondingly, the saturation of each pore phase is expressed as $s_\beta = V_\beta/(\phi V)$ and satisfies $\sum s_\beta = 1$. According to the mixture theory, the partial densities associated with each phase are given by:

$$\widetilde{\rho}_s = (1-\phi)\rho_s, \quad \widetilde{\rho}_\beta = \phi s_\beta \rho_\beta\,, \tag{2}$$

where ρ_s and ρ_β represent the intrinsic densities of solid phase s and pore phase β, respectively.

2.2. Water and gas flow

The seepage behaviors of multiphase fluids (e.g., water and gas) in a porous medium are characterized by the generalized Darcy's law, which are mathematically expressed as:

$$\begin{aligned} \boldsymbol{q}_w &= \phi s_w(\boldsymbol{v}_w - \boldsymbol{v}_s) = -\frac{k_{rw}\boldsymbol{k}}{\mu_w}(\nabla p_{\mathrm{w}} - \rho_{\mathrm{w}}\boldsymbol{g}), \\ \boldsymbol{q}_g &= \phi s_g(\boldsymbol{v}_g - \boldsymbol{v}_s) = -\frac{k_{rg}\boldsymbol{k}}{\mu_g}\left(\nabla p_g - \rho_g \boldsymbol{g}\right). \end{aligned} \tag{3}$$

In Eq. (3), the subscripts w and g denote the parameters related to the water and gas phases. $\boldsymbol{q}$, p and μ represent the Darcy velocity vector, pressure and dynamic viscosity, respectively. Under the assumption of isotropy, the absolute permeability matrix $\mathbf{k}$ is given by the following relationship:

$$\boldsymbol{k} = k_h \boldsymbol{I}\,, \tag{4}$$

in which k_h represents the intrinsic permeability measured from the experimental data. To describe the permeability evolution during hydrate dissociation, the coefficient k_h is expressed by the function $k_h = k_{h0}(1-s_h)^N$, with k_{h0} representing the initial intrinsic permeability in the hydrate-free condition and N denoting an empirical exponent that quantifies the permeability reduction. The relative permeabilities of water (k_{rw}) and gas (k_{rg}) are introduced as functions of their respective saturations to quantify the multi-phase flow behavior in porous media. For modeling methane hydrate dissociation, the relative permeabilities are formulated using a modified Corey correlation (Castaneda et al., 2025; Liang et al., 2010; Ruan et al., 2012) as follows:

$$\begin{aligned} k_{rw} &= \left(\frac{s_w/(s_w+s_g) - s_{wr}}{1 - s_{wr} - s_{gr}}\right)^4, \\ k_{rg} &= \left(\frac{s_g/(s_w+s_g) - s_{gr}}{1 - s_{wr} - s_{gr}}\right)^2, \end{aligned} \tag{5}$$

where the residual saturations of water (s_{wr}) and gas (s_{gr}) are accounted for. For unsaturated soils without considering hydrate dissociation, various relative permeability models can be found in Refs (Mualem, 1976; Leong and Rahardjo, 1997; Rahimi et al., 2015). In saturated soils where the pore space is exclusively filled with water, the relative permeability coefficient of water is taken as 1.0, as it experiences no interference from other fluid phases.

2.3. Mass conservation equation

To accurately capture interphase mass transfer and phase transitions, separate mass conservation equations are required for each phase constituting the porous medium. Within the current computational framework, the mass conservation of the solid is employed to explicitly update the porosity, while the saturations of the methane hydrate and water are explicitly calculated via their respective mass conservation equations. By combining the mass conservation equations for gas and water, a mixed mass conservation equation for the fluid phases is derived, which provides the theoretical basis for the implicit solution of the pressure field.

1) Solid grains.

Although the intrinsic density of solid grains exhibits dependence on both confining pressure and temperature, the influence of confining pressure is considered negligible, as noted by Zhao and Borja (Zhao and Borja, 2020). Consequently, the solid density is assumed to be temperature-dependent only, which is expressed as:

$$\rho_s = \rho_s^0 \exp[-3\alpha_s(T - T_0)], \tag{6}$$

in which ρ_s^0 represents the constant solid density under the reference temperature T_0. α_s stands for the solid's linear thermal expansion coefficient, while T denotes the current temperature. Neglecting the mass exchange between the solid phase and others, the mass conservation equation for solid grains, excluding the source term, is described as:

$$(1-\phi)\nabla\cdot\boldsymbol{v}_s - \frac{\mathrm{d}\phi}{\mathrm{d}t} - 3\alpha_s(1-\phi)\frac{\mathrm{d}T}{\mathrm{d}t} = 0. \tag{7}$$

Eq. (7) presents the final form of the mass conservation equation for solid grains, with its complete derivation detailed in Appendix-A. Utilizing the relation $\nabla\cdot\boldsymbol{v}_s = \mathrm{d}\varepsilon_v/\mathrm{d}t$ and further rearranging Eq. (7), the porosity increment is derived as follows:

$$\mathrm{d}\phi = (1-\phi)\mathrm{d}\varepsilon_v - 3\alpha_s(1-\phi)\mathrm{d}T. \tag{8}$$

2) Methane hydrate.

In a porous medium containing methane hydrate, fluctuations in pressure and temperature may induce a phase transition that results in either the dissociation or synthesis of methane hydrate, as illustrated by the following reaction:

$$CH_4\cdot nH_2O(s) \underset{\text{Formation}}{\overset{\text{Dissociation}}{\longleftrightarrow}} CH_4(g) + nH_2O(w) \quad . \tag{9}$$

The mass dissociation rates of each component in kinetics reaction are expressed as follows:

$$\dot{m}_h = -M_h R_h, \quad \dot{m}_w = n\cdot M_w R_h, \quad \dot{m}_g = M_g R_h, \tag{10}$$

where $M_\beta\ (\beta = w, g, h)$ is the molar mass of phase β, and n denotes the hydration number. Based on the Kim-Bishnoi kinetic model (Kim et al., 1987), the dissociation rate R_h is defined as:

$$R_h = k_{d0}\exp\left(-\frac{\Delta E}{RT}\right)A_s\left(p_{eq} - p_g\right), \tag{11}$$

in which k_{d0}, ΔE and R are the intrinsic dissociation rate constant, activation energy and gas constant, respectively. The interface area A_s is defined as suggested by White et al. (2020):

$$A_s = \begin{cases} \phi s_h A_{geo}, & \text{if } p \le p_{eq} \\ s_w s_g (1.0 - s_h) A_{geo}, & \text{if } p > p_{eq} \end{cases}, \tag{12}$$

where A_{geo} stands for the geometric ratio of surface area versus volume, derived from the particle diameter. Additionally, the methane equilibrium pressure p_{eq} is determined using an empirical formula (Clarke and Bishnoi, 2000; Sun et al., 2005):

$$p_{eq} = 1.15\exp(49.3185 - 9459/T). \tag{13}$$

Given that the solid grains and hydrate exhibit no relative motion (i.e., $v_h = v_s$), the mass conservation equation for the hydrate can be readily derived as follows:

$$\begin{aligned} &\frac{\mathrm{d}(\phi s_h \rho_h)}{\mathrm{d}t} + \phi s_h \rho_h \nabla\cdot\boldsymbol{v}_h = -M_h R_h \\ \Rightarrow &\frac{\mathrm{d}s_h}{\mathrm{d}t} = -\frac{M_h R_h}{\rho_h \phi} - \frac{s_h}{\phi}\dot{\varepsilon}_v + 3\alpha_s(1-\phi)\frac{s_h}{\phi}\frac{\mathrm{d}T}{\mathrm{d}t}. \end{aligned} \tag{14}$$

With the detailed derivation of Eq. (14) outlined in Appendix-A, the increment of hydrate saturation is further determined by:

$$\mathrm{d}s_h = -\frac{M_h R_h}{\phi\rho_h}\mathrm{d}t - \frac{s_h}{\phi}\mathrm{d}\varepsilon_v + 3\alpha_s(1-\phi)\frac{s_h}{\phi}\mathrm{d}T\,. \tag{15}$$

3) Water.

Treating water as an incompressible fluid allows for the straightforward derivation of the mass conservation equation shown below:

$$\begin{aligned} &\frac{\partial(\phi s_w \rho_w)}{\partial t} + \nabla\cdot(\phi s_w \rho_w \boldsymbol{v}_w) = 6M_w R_h \\ \Rightarrow &\phi\rho_w\frac{\mathrm{d}s_w}{\mathrm{d}t} + s_w\rho_w\frac{\mathrm{d}\phi}{\mathrm{d}t} + \phi_w\rho_w\nabla\cdot\boldsymbol{v}_s + \rho_w\nabla\cdot\boldsymbol{q}_w = 6M_w R_h\,. \end{aligned} \tag{16}$$

By substituting Eq. (7) into Eq. (16), an explicit expression for updating water saturation is derived as:

$$\mathrm{d}s_w = \frac{6M_w R_h}{\phi\rho_w}\mathrm{d}t - \frac{s_w}{\phi}\mathrm{d}\phi - s_w\mathrm{d}\varepsilon_v - \frac{\nabla\cdot\boldsymbol{q}_w}{\phi}\mathrm{d}t\ . \tag{17}$$

4) Gas

The gas saturation increment can be deduced by applying the conservation law, provided that the increments of hydrate and water saturation have been solved from Eqs. (15) and (17):

$$\mathrm{d}s_g = -\mathrm{d}s_h - \mathrm{d}s_w\,. \tag{18}$$

Once the gas saturation is known, the mass conservation equation for the gas phase is subsequently established as:

$$\begin{aligned} &\frac{\partial(\phi s_g \rho_g)}{\partial t} + \nabla\cdot(\phi s_g \rho_g \boldsymbol{v}_g) = M_g R_h \\ \Rightarrow &\phi\frac{\mathrm{d}s_g}{\mathrm{d}t} + s_g\frac{\mathrm{d}\phi}{\mathrm{d}t} + \frac{\phi s_g}{\rho_g}\frac{\mathrm{d}\rho_g}{\mathrm{d}t} + \phi s_g\nabla\cdot\boldsymbol{v}_s + \nabla\cdot\boldsymbol{q}_g + \frac{\nabla\rho_g}{\rho_g}\boldsymbol{q}_g = \frac{M_g R_h}{\rho_g}. \end{aligned} \tag{19}$$

The detailed derivation of Eq. (19) can be found in Appendix-A. For an ideal gas, the derivation based on its equation of state is presented below:

$$\rho_g = \frac{M_g p_g}{RT} \quad \Rightarrow \quad \frac{\mathrm{d}\rho_g}{\mathrm{d}t} = \frac{M_g\cdot\mathrm{d}p_g}{RT\cdot\mathrm{d}t} - \frac{M_g p_g\cdot\mathrm{d}T}{RT^2\cdot\mathrm{d}t} = \frac{\rho_g}{p_g}\dot{p}_g - \frac{\rho_g}{T}\dot{T}\,. \tag{20}$$

By combining Eqs. (7), (16), (19), and (20), we derive the hybrid mass conservation equation for the water–gas system:

$$\begin{aligned} &\frac{(s_g + s_w)}{\phi}\dot{\varepsilon}_v + \frac{s_g}{p_g}\dot{p}_g + \frac{\nabla\rho_g}{\phi\rho_g}\boldsymbol{q}_g + \frac{\left(\nabla\cdot\boldsymbol{q}_g + \nabla\cdot\boldsymbol{q}_w\right)}{\phi} - \frac{s_g}{T}\dot{T} \\ &- \frac{3\alpha_s(s_g + s_w)(1-\phi)}{\phi}\dot{T} \\ &= \dot{s}_h + \frac{M_g R_h}{\phi\rho_g} + \frac{6M_w R_h}{\phi\rho_w}. \end{aligned} \tag{21}$$

The full mathematical derivation regarding the conservation of mass for the hybrid water–gas system is presented in Appendix-A. Once the pressure of either the water or gas phase is determined using Eq. (21), the pressure of the other phase can be calculated using the following empirical capillary pressure p_c expression:

$$p_c = p_g - p_w = p_c^e\left(\frac{s_w/(s_w + s_g) - s_{wr}}{1 - s_{wr}}\right)^{-0.65}. \tag{22}$$

2.4. Momentum conservation equation

In quasi-static analyses where inertial effects are disregarded, the momentum balance for solid grains, as recommended in Ref (Tang et al., 2026), is formulated as follows:

$$\begin{aligned}\nabla\cdot\boldsymbol{\sigma}+\boldsymbol{b}&=\nabla\cdot\left(\boldsymbol{\sigma}^{'}-\frac{s_w p_w+s_g p_g}{s_w+s_g}\boldsymbol{I}\right)+\boldsymbol{b}\\&=\nabla\cdot\left(\boldsymbol{\sigma}^{'}-p_w\boldsymbol{I}-\frac{s_g}{s_w+s_g}p_c\boldsymbol{I}\right)+\boldsymbol{b}=0\,.\end{aligned}\tag{23}$$

in which the identity matrix is denoted by **I** and the body force vector is represented by **b**. Furthermore, the total stress is indicated as $\boldsymbol{\sigma}$, with the effective stress denoted as $\boldsymbol{\sigma}'$, whose evolution is governed by elasto-plastic and thermal strains, as defined by:

$$\mathrm{d}\boldsymbol{\sigma}'=\mathbb{D}^{ep}:(\mathrm{d}\boldsymbol{\varepsilon}-\mathrm{d}\boldsymbol{\varepsilon}_T)\,.\tag{24}$$

In Eq. (24), the term $\mathbb{D}^{ep}$ refers to the constitutive tensor, with $\boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}_T$ corresponding to the solid's total strain and thermal strain, calculated via:

$$\boldsymbol{\varepsilon}=\left(\nabla\boldsymbol{u}+(\nabla\boldsymbol{u})^T\right)/2,\qquad\boldsymbol{\varepsilon}_T=\alpha_s(T-T_0)\boldsymbol{I}\ .\tag{25}$$

where $\boldsymbol{u}$ denotes the displacement vector. For methane hydrate-bearing soils, an empirical model connecting Young's modulus to hydrate saturation was proposed by Klar et al. (2010) based on triaxial testing results, as shown below:

$$E=(300+1350s_h)\,\mathrm{MPa}\ .\tag{26}$$

2.5. Energy conservation equation

In this study, heat transfer is modeled primarily through conduction and convection. Thermal radiation is neglected as its contribution is considered negligible within the porous media (Tang et al., 2024), allowing for a more focused and efficient computational approach. Dimensional analysis by Gelet et al. (2012) indicates that, compared to heat conduction, the thermal effects induced by hydraulic-mechanical coupling are negligible. In accordance with the theoretical frameworks delineated in Refs (Li et al., 2024; Tang et al., 2024; Wu et al., 2025); the REV energy conservation equation is derived by omitting hydraulic-mechanical coupling effects, as follows:

$$\rho c\partial T\big/\partial t=-\nabla\cdot\boldsymbol{Q}_d-\left(\rho_w c_w\boldsymbol{q}_w+\rho_g c_g\boldsymbol{q}_g\right)\cdot\nabla T+Q_T\,,\tag{27}$$

where, ρc is the effective volumetric heat capacity of porous media, and $\boldsymbol{Q}_d$ represents the thermal conduction term. Q_T denotes the heat source and equals zero in porous media without hydrates or external heating; For hydrate decomposition, Q_T represents the heat source term arising from phase transition. The components in Eq. (27) are defined mathematically as follows:

$$\begin{cases}\rho c=(1-\phi)\rho_s c_s+s_h\phi\rho_h c_h+s_w\phi\rho_w c_w+s_g\phi\rho_g c_g\\ \boldsymbol{Q}_d=-k_T\nabla T=-\left((1-\phi)k_{Ts}+s_h\phi k_{Th}+s_w\phi k_{Tw}+s_g\phi k_{Tg}\right)\nabla T\\ Q_T=-M_h R_h\Delta H\end{cases}\ .\tag{28}$$

In Eq. (28), $\rho_\alpha c_\alpha$ and $k_{T\alpha}$ are the heat capacity and of thermal conductivity coefficient of phase $\alpha(e.g.\ ,\quad \alpha=s,h,w,g)$, respectively. ΔH quantifies the enthalpy change due to the latent heat of hydrate decomposition and is considered as a linear function of temperature (Sellm and Sloan, 1990):

$$\Delta H=446120-132.638T\,.\tag{29}$$

2.6. Weak forms of conservation equations

Within the multi-physics coupling framework that accounts for hydrate chemical reactions, the system comprises seven unknown variables. Among these variables, porosity, hydrate saturation, water saturation, and gas saturation are computed explicitly, whereas the displacement, pressure, and temperature fields are solved implicitly. This subsection focuses on deriving the weak forms for the conservation equations of momentum, mass, and energy, which correspond to the fields of displacement, pressure, and temperature.

1) Weak forms governing momentum conservation

Using standard Galerkin discretization with the displacement field $\delta\mathbf{u}$ as the test function, we arrive at the weak form of the momentum conservation equation for solids:

$$\begin{aligned}\mathscr{L}_u=&\int_\Omega\nabla\delta\boldsymbol{u}:\boldsymbol{\sigma}^{'}\mathrm{d}\Omega-\int_\Omega p_w\nabla\cdot\delta\boldsymbol{u}\mathrm{d}\Omega\\&-\int_\Omega\frac{s_g}{s_w+s_g}p_c\nabla\cdot\delta\boldsymbol{u}\mathrm{d}\Omega-\int_\Gamma\delta\boldsymbol{u}\cdot\bar{\boldsymbol{t}}\mathrm{d}\Gamma-\int_\Omega\delta\boldsymbol{u}\cdot\boldsymbol{b}\mathrm{d}\Omega\,.\end{aligned}\tag{30}$$

In the configuration Ω, the displacement field satisfies the Dirichlet boundary conditions Γ_D, while the stress follows the Neumann boundary conditions Γ_N, as specified below:

$$\begin{cases}\boldsymbol{u}=\bar{\boldsymbol{u}} & \text{on}\ \Gamma_D\\ \boldsymbol{n}\cdot\boldsymbol{\sigma}=\bar{\boldsymbol{t}} & \text{on}\ \ \Gamma_N\end{cases}\ .\tag{31}$$

In Eq. (31), $\bar{\boldsymbol{t}}$ denotes the surface traction corresponding to the total stress, while $\boldsymbol{n}$ represents the unit normal vector pointing outward from the Neumann boundary.

2) Weak forms governing mass conservation

According to the mass conservation equation (21) of the hybrid gas–water system, we employ the standard Galerkin discretization with the variation of water pressure δp_w as the test function to derive its weak form:

$$\begin{aligned}\mathscr{L}_p=&\int_\Omega\delta p_w\frac{(s_g+s_w)}{\phi}\dot{\varepsilon}_v\mathrm{d}\Omega+\int_\Omega\delta p_w\frac{s_g}{\rho_g}\frac{M_g}{RT}(\dot{p}_w+\dot{p}_c)\mathrm{d}\Omega\\&+\int_\Omega\delta p_w\frac{\boldsymbol{q}_g^T}{\phi\rho_g}\frac{M_g}{RT}(\nabla p_w+\nabla p_c)\mathrm{d}\Omega+\int_\Gamma\delta p_w\frac{\bar{q}_g}{\phi}\mathrm{d}\Gamma+\int_\Gamma\delta p_w\frac{\bar{q}_w}{\phi}\mathrm{d}\Gamma\\&-\int_\Omega\nabla(\delta p_w)\frac{\boldsymbol{q}_g}{\phi}\mathrm{d}\Omega-\int_\Omega\nabla(\delta p_w)\frac{\boldsymbol{q}_w}{\phi}\mathrm{d}\Omega\\&-\int_\Omega\delta p_w\left(\dot{s}_h+\frac{M_g R_h}{\phi\rho_g}+\frac{6M_w R_h}{\phi\rho_w}\right)\mathrm{d}\Omega\\&-\int_\Omega\delta p_w\left(\frac{s_g}{T}+\frac{3\alpha_s(s_g+s_w)(1-\phi)}{\phi}\right)\dot{T}\mathrm{d}\Omega-\int_\Omega\delta p_w\frac{\boldsymbol{q}_g^T}{\phi T}\nabla T\mathrm{d}\Omega\,.\end{aligned}\tag{32}$$

In Eq. (32), the conditions for pressure and flux boundaries adhere to the expressions below:

$$\begin{cases}p=\bar{p} & \text{on}\ \Gamma_p\\ \boldsymbol{n}\cdot\boldsymbol{q}=\bar{q} & \text{on}\ \Gamma_q\end{cases}\tag{33}$$

in which, $\bar{p}$ denotes the water pressure prescribed on the Dirichlet boundary Γ_p, and $\bar{q}$ represents the normal flux that satisfies the Neumann boundary Γ_q.

3) Weak forms governing energy conservation

Multiplying Eq. (27) by a test function δT and applying the divergence theorem, the weak form of the energy conservation equation is derived as follows:

$$\begin{aligned}\mathscr{L}_T &= \int_\Omega \delta T \rho c \dot{T} \mathrm{d}\Omega + \int_\Omega \delta T \left(\rho_w c_w \boldsymbol{q}_w + \rho_g c_g \boldsymbol{q}_g\right) \nabla T \mathrm{d}\Omega \\ &\quad + \int_\Omega \delta T \nabla \cdot \boldsymbol{Q}_d \mathrm{d}\Omega + \int_\Omega \delta T \cdot M_h R_h \Delta H \mathrm{d}\Omega \\ &= \int_\Omega \delta T \rho c \dot{T} \mathrm{d}\Omega + \int_\Omega \delta T \left(\rho_w c_w \boldsymbol{q}_w + \rho_g c_g \boldsymbol{q}_g\right) \nabla T \mathrm{d}\Omega \\ &\quad - \int_{\Gamma_D} \underbrace{\delta T}_{=0} Q_D \mathrm{d}\Gamma - \int_{\Gamma_N} (\delta T) Q_N \mathrm{d}\Gamma + \int_{\Gamma_R} (\delta T) h_r (T - T_r)\, \mathrm{d}\Gamma \\ &\quad + \int_\Omega \nabla(\delta T) k_T \nabla T \mathrm{d}\Omega + \int_\Omega \delta T \cdot M_h R_h \Delta H \mathrm{d}\Omega \,.\end{aligned} \tag{34}$$

Eq. (34) describes a heat transfer problem that involves three types of boundary conditions, namely the Dirichlet, Neumann, and Robin boundaries, which are given by the following expressions:

$$\begin{cases} T = T_d & (\text{on } \Gamma_D) \\ \boldsymbol{n}\cdot\boldsymbol{q} = Q_N = \overline{q}^n & (\text{on } \Gamma_N) \\ \boldsymbol{n}\cdot\boldsymbol{q} = h_r(T - T_r) = \overline{q}^r & (\text{on } \Gamma_R) \end{cases} . \tag{35}$$

In Eq. (35), T_d and $\overline{q}^n$ refer to the imposed temperature on the Dirichlet boundary and the heat flux on the Neumann boundary, respectively. h_r characterizes the coefficient governing convective heat transfer, and T_r denotes the ambient temperature.

2.7. Spatial and temporal discretization

In classical FE schemes, the displacement field is typically discretized with higher-order shape functions than those used for the pressure field. This strategy is essential for satisfying the inf-sup stability criterion (Babuška and Narasimhan, 1997) and mitigating spurious pressure oscillations. The bES-FEM incorporates bubble nodes to satisfy the inf-sup condition effectively. This allows for the use of efficient low-order elements to discretize the displacement, pressure, and temperature fields without suffering from pressure instabilities. The corresponding interpolation expressions are provided below:

$$\boldsymbol{u} = \boldsymbol{N}_u \boldsymbol{U}, \quad p_w = \boldsymbol{N}_p \boldsymbol{P}, \quad T = \boldsymbol{N}_T \boldsymbol{T}, \quad \phi = \boldsymbol{N}_\phi \boldsymbol{\Phi} \,, \tag{36}$$

in which, $\boldsymbol{U}$, $\boldsymbol{P}$, $\boldsymbol{T}$ and $\boldsymbol{\Phi}$ are vectors of nodal displacement, water pressure, temperature and porosity, respectively. $\boldsymbol{N}_\alpha\,(\alpha = u, p, T, \phi)$ denotes the shape function corresponding to physical field α. Substituting Eq. (36) into Eqs. (30),(32) and (34), the matrix formulations of the spatially discretized equations can be derived as follows:

$$\boldsymbol{K}_u \boldsymbol{U} + \boldsymbol{K}_{uw} \boldsymbol{P} + \boldsymbol{K}_{uT} \boldsymbol{T} = \boldsymbol{F}^u \,, \tag{37}$$

$$\boldsymbol{K}_{wu} \dot{\boldsymbol{U}} + \boldsymbol{K}_{ww} \dot{\boldsymbol{P}} + \boldsymbol{K}_w \boldsymbol{P} + \boldsymbol{K}_{wTT} \dot{\boldsymbol{T}} + \boldsymbol{K}_{wT} \boldsymbol{T} = \boldsymbol{F}^w \,, \tag{38}$$

$$\boldsymbol{K}_{TT} \dot{\boldsymbol{T}} + \boldsymbol{K}_T \boldsymbol{T} = \boldsymbol{F}^h \,, \tag{39}$$

where detailed derivations of the individual terms comprising the stiffness matrix are presented in Appendix-B.

For temporal discretization, the present research adopts a scheme derived from the trapezoidal rule (Tang et al., 2024; Sloan and Abbo, 1999); which assumes that physical quantities evolve linearly between the current time step t_n and next time step t_{n+1}. Consequently, the physical variables at time $t_{n+\theta}$ for both hybrid mass and energy conservation are approximated via linear interpolation, as expressed below:

$$\begin{cases} \boldsymbol{P}_{t_{n+\theta}} = (1-\theta_1)\boldsymbol{P}_{t_n} + \theta_1 \boldsymbol{P}_{t_{n+1}}, & \boldsymbol{F}^p_{t_{n+\theta}} = (1-\theta_1)\boldsymbol{F}^p_{t_n} + \theta_1 \boldsymbol{F}^p_{t_{n+1}} \\ \boldsymbol{T}_{t_{n+\theta}} = (1-\theta_2)\boldsymbol{T}_{t_n} + \theta_2 \boldsymbol{T}_{t_{n+1}}, & \boldsymbol{F}^h_{t_{n+\theta}} = (1-\theta_2)\boldsymbol{F}^h_{t_n} + \theta_2 \boldsymbol{F}^h_{t_{n+1}} \end{cases} . \tag{40}$$

Typically, the implicit solution is guaranteed to be stable and convergent provided that the θ value is maintained within the range [0.5, 1.0]. In the present study, the parameter θ is prescribed as 1, thereby corresponding to the backward Euler formulation. By applying the time discretization scheme in Eq. (40) to Eqs. (38)~(39) and subsequently integrating over the designated time increment Δt, we obtain the following expressions:

$$\begin{aligned}&\boldsymbol{K}_{wu}\boldsymbol{U}_{t_{n+1}} + (\boldsymbol{K}_{ww} + \Delta t \boldsymbol{K}_w)\boldsymbol{P}_{t_{n+1}} + (\boldsymbol{K}_{wTT} + \Delta t \boldsymbol{K}_{wT})\boldsymbol{T}_{t_{n+1}} \\ &\quad = \Delta t \boldsymbol{F}^w_{t_{n+1}} + \boldsymbol{K}_{wu}\boldsymbol{U}_{t_n} + \boldsymbol{K}_{ww}\boldsymbol{P}_{t_n} + \boldsymbol{K}_{wTT}\boldsymbol{T}_{t_n} \,,\end{aligned} \tag{41}$$

$$\boldsymbol{K}_{TT}\boldsymbol{T}_{t_{n+1}} + \Delta t \boldsymbol{K}_T \boldsymbol{T}_{t_{n+1}} = \Delta t \boldsymbol{F}^h_{t_{n+1}} + \boldsymbol{K}_{TT}\boldsymbol{T}_{t_n} \,. \tag{42}$$

Combining Eq. (37), (41), and (42), the matrix formulation for implicitly solving the displacement, pore water pressure, and temperature fields can be derived as follows:

$$\begin{aligned}&\begin{bmatrix} \boldsymbol{K}_u & \boldsymbol{K}_{uw} & \boldsymbol{K}_{uT} \\ \boldsymbol{K}_{wu} & \boldsymbol{K}_{ww} + \Delta t \boldsymbol{K}_w & \boldsymbol{K}_{wTT} + \Delta t \boldsymbol{K}_{wT} \\ 0 & 0 & \boldsymbol{K}_{TT} + \Delta t \boldsymbol{K}_T \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{t_{n+1}} \\ \boldsymbol{P}_{t_{n+1}} \\ \boldsymbol{T}_{t_{n+1}} \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{F}^u_{t_{n+1}} \\ \Delta t \boldsymbol{F}^w_{t_{n+1}} + \boldsymbol{K}_{wu}\boldsymbol{U}_{t_n} + \boldsymbol{K}_{ww}\boldsymbol{P}_{t_n} + \boldsymbol{K}_{wTT}\boldsymbol{T}_{t_n} \\ \Delta t \boldsymbol{F}^h_{t_{n+1}} + \boldsymbol{K}_{TT}\boldsymbol{T}_{t_n} \end{bmatrix} .\end{aligned} \tag{43}$$

In addition, the explicit update of saturation imposes a strict stability constraint on the time step, known as the CFL condition (Courant et al., 1967). This condition is determined by the maximum eigenvalue of the system's Jacobian matrix, which corresponds to the speed of the fastest propagating saturation wave. In a three-phase system (water, gas, and hydrate), the hydrate phase is typically assumed to have limited mobility. Therefore, the focus is primarily on the flows of the water (w) and gas (g) phases. For water–gas two-phase flow, the generalized CFL condition is formulated as follows:

$$\text{CFL:} \quad \frac{\Delta t}{\Delta x} \cdot \max\left(\left\| \frac{\partial \boldsymbol{q}_w}{\partial s_w} \right\|, \left\| \frac{\partial \boldsymbol{q}_g}{\partial s_g} \right\| \right) \leqslant 1 \,, \tag{44}$$

in which Δt and Δx denote the time step and characteristic element size. According to the methodologies in Refs (Coats, 2003a,b), we adopt the following modified CFL criterion:

$$\Delta t_{CFL} \leqslant \min_{e\in\Omega} \left(\frac{\phi(1.0 - s_h)\cdot \Delta x}{\|\boldsymbol{q}_t\| \cdot \max\left(f'_w(s_w), f'_g(s_g)\right)} \right) , \tag{45}$$

where $\boldsymbol{q}_t$ denotes the total flow rate of the water and gas phases (i.e., $\boldsymbol{q}_t = \boldsymbol{q}_w + \boldsymbol{q}_g$).$f'_\alpha(s_\alpha)$ represents the derivative of the fractional flow function with respect to the saturation for phase α, where α corresponds to either water (w) or gas (g). For saturated soils without hydrates, the CFL condition in Eq. (45) can be further simplified to $\Delta t_{CFL} \leqslant \min_{e\in\Omega}(\phi \cdot \Delta x / \|\boldsymbol{q}_w\|)$, which is consistent with the recommendations in Refs (Li et al., 2024; Coats, 2003b).

Notably, hydrate systems involve not only fluid flow but also intense phase transitions. If the hydrate dissociation proceeds too rapidly within a single time step, the hydrate saturation may become negative or oscillatory. Consequently, stricter time step constraints are required, as follows:

$$\Delta t_{sh} \leqslant \min_{e\in\Omega}(\eta \cdot s_h / |\dot{s}_h|) \,, \tag{46}$$

in which η is the safety reduction factor less than 1.0. A fixed value of ($\eta = 0.5$) was used for all simulation cases in this study through extensive repeated numerical testing, yielding relatively stable computational performance. A smaller η gives a more conservative time step and improves robustness at the expense of computational efficiency, whereas an excessively large η may lead to negative or oscillatory hydrate saturation in rapidly dissociating regions. Notably, η only controls the admissible time-step size and does not modify the governing equations, hydrate dissociation kinetics, material parameters, or boundary conditions. To ensure computational robustness, the final explicit time step Δt is determined using the following combined constraints:

$$\Delta t = \min(\Delta t_{CFL}, \Delta t_{sh}) \,. \tag{47}$$

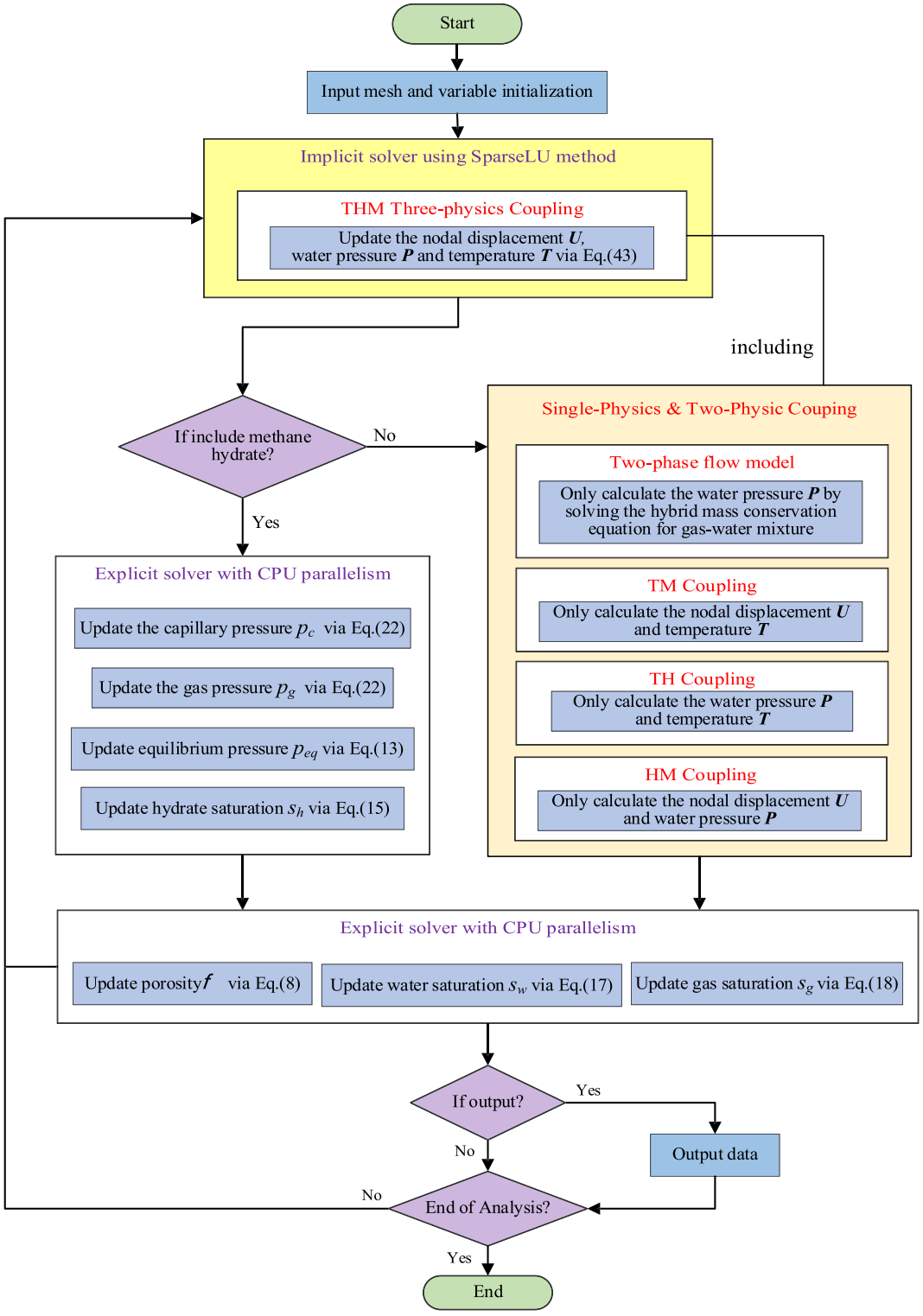

Fig. 2. Schematic of the CPU-Parallel SparseLU strategy within the unified multi-physics coupled framework.

3. Numerical implementation strategy

The proposed numerical implementation strategy is motivated by addressing the potential numerical challenges arising from hydrate dissociation. During depressurization or thermal stimulation, hydrate decomposition generates coupled variations in pore pressure, phase saturation, temperature, permeability, and sediment stiffness. These processes give rise to two major numerical challenges. On the one hand, low permeability and near undrained conditions may induce a nearly incompressible mechanical response, which can lead to spurious pore pressure oscillations and volumetric locking in conventional finite element formulations. On the other hand, the release and migration of gas and water, together with heat transport, often produce convection dominated mass and thermal fronts that are prone to nonphysical oscillations. To address these challenges while maintaining consistency with the underlying hydrate dissociation mechanisms, the proposed

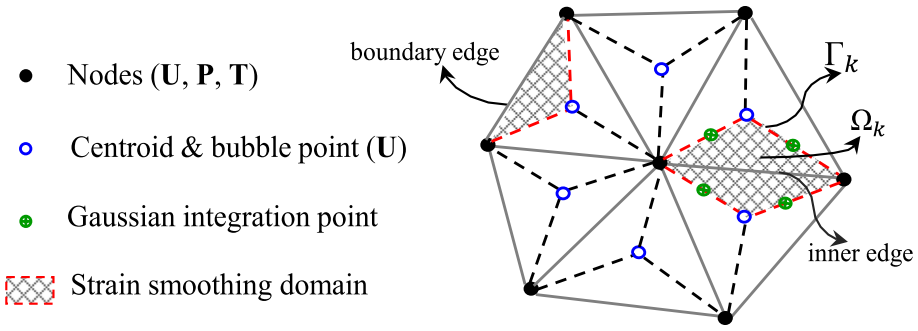

Fig. 3. Schematic illustration of the geometry and integration scheme for the bES-FEM.

framework integrates bES-FEM and SUPG into a unified stabilization strategy. The bES-FEM formulation enhances the stability of the displacement pressure coupling under near incompressible conditions, while the SUPG formulation provides streamline oriented stabilization for convection dominated transport processes. Through the integration of these methods, the proposed framework guarantees the stability, accuracy, and efficiency required for complex multi-physics simulations.

3.1. Unified framework for multi-physics coupling calculation

In the current unified framework, the displacement, water pressure and temperature fields are implicitly solved, while other physical quantities, such as saturation and porosity, are explicitly updated. This explicit–implicit partition is adopted as a practical and widely used numerical treatment for strongly nonlinear coupled problems. The advantage of the present framework lies in embedding this treatment into a unified computational architecture that can activate or suppress different governing equations according to the considered coupling mode. As shown in Fig. 2, it is applicable not only to fully coupled THMC processes involving hydrate dissociation but also to three-field THM coupling where phase changes induced by hydrate decomposition are absent. Moreover, it is compatible with simplified two-field or single-field coupling models. Within the current framework, the basic strategy for solving three-field, two-field, or single-field coupling models is to set the saturation of omitted phases to zero and assume constant values for the neglected physical fields, thereby eliminating the need to solve the corresponding governing equations. For example, when dealing with THM coupling in saturated soils, the porous medium is considered fully water-saturated (with s_w fixed at 1.0) while the hydrate and gas phases are assumed absent (with saturations fixed at 0). This treatment strategy bypasses the need to account for intermediate variables associated with hydrate decomposition. Furthermore, when variations in the temperature within saturated soils are no longer of primary interest, the THM coupling model can be naturally simplified to the HM coupling model. Similarly, by neglecting solid deformation or assuming constant pore water pressure, the corresponding TH and TM coupling models can be derived. For problems involving the displacement of gas and water in unsaturated soils, it is sufficient to obtain the water pressure by implicitly solving the hybrid mass conservation equation for the gas–water mixture. Subsequently, the saturations of the water and gas phases within the pore space are explicitly updated, thereby completing the numerical solution for the two-phase flow problem. Thus, the current unified multi-physics framework can be regarded as demonstrating high compatibility and versatility.

Notably, the current unified high-compatibility frameworks provide essential capabilities for modeling complex geological conditions in geotechnical and energy engineering. Taking deep-sea methane hydrate exploitation as an example, the process involves significant spatial heterogeneity, necessitating differentiated coupling strategies across different regions. Specifically, the hydrate reservoir layer requires a fully coupled THMC analysis, whereas the hydrate-free overlying and underlying saturated soil layers involve only THM coupling. Furthermore, the shallow seawater region can be simplified to a TH coupling problem. While coordinating multiple coupling schemes within a single model has traditionally been a formidable task, achieving spatially differentiated multi-physics coupling poses an even more significant challenge. Fortunately, the unified framework introduced herein offers a viable solution to this complexity, enabling the simulation of such intricate conditions. Detailed application examples are presented in Section 5.

In addition, complex multi-physics problems often involve extensive and intricate computations, which results in significant computational overhead. However, the current framework offers an additional advantage: the self-developed platform implements CPU parallelization (via OpenMP) for explicit solvers and utilizes sparse matrix techniques (via the Eigen library) for implicit solutions, substantially reducing computational costs. All numerical simulations were performed on a personal computer equipped with an Intel Core i9-14900KF CPU (3.2 GHz) and 32 GB of RAM.

3.2. Enhance edge-based smoothing via bubble functions

In contrast to traditional FEM, the ES-FEM partitions each element into multiple smoothing domains and applies boundary integrals to average the strain field, effectively reducing local strain peaks and inducing the "softening effects" (Tang et al., 2024). This results in numerical solutions that are more accurate and robust against mesh distortions. Although ES-FEM demonstrates substantial potential, it does not entirely overcome volumetric locking in nearly incompressible materials (Nguyen-Xuan and Liu, 2013). To address this limitation, the T3 element is enriched with a centrally positioned bubble node to form a MINI element (Arnold et al., 1984; Mo et al., 2025), within which an edge-based smoothed integration technique is applied to develop the bES-FEM formulation. The bES-FEM not only exhibits enhanced "softening effects" but has also been validated to satisfy the inf-sup condition (Ho et al., 2025), thereby demonstrating exceptional stability and broad applicability for solving problems in the incompressible limit. After incorporating the bubble node, the displacement **u** within the T3 element is expressed using the following interpolation function:

$$\boldsymbol{u} = \sum_{i=1}^{3} \begin{bmatrix} N_u^i(\boldsymbol{x}) & 0 \\ 0 & N_u^i(\boldsymbol{x}) \end{bmatrix} \boldsymbol{U}_i + \begin{bmatrix} N_u^b(\boldsymbol{x}) & 0 \\ 0 & N_u^b(\boldsymbol{x}) \end{bmatrix} \boldsymbol{U}_b \ , \tag{48}$$

in which $\boldsymbol{U}_i$ denotes the displacement vectors at the three nodes of the T3 element, and $\boldsymbol{U}_b$ represents the displacement vector of the bubble node. $N_u^i(\boldsymbol{x})$ is the shape function value at coordinates **x** for the i th node of the T3 element, while $N_u^b(\boldsymbol{x})$ denotes the shape function value at point **x** for the bubble node, expressed as follows:

$$N_u^b = 27\lambda_1\lambda_2\lambda_3 \quad (\lambda_i = a_i + b_i x + c_i y;\ i = 1,2,3)\ . \tag{49}$$

In Eq. (49), λ_i denotes the area coordinate associated with the

coefficients a_i, b_i, and c_i, which satisfy the following expressions:

$$a_i = \left(x_j y_k - x_k y_j\right)\Big/2A_e, \quad b_i = \left(y_j - y_k\right)\Big/2A_e, \quad c_i = \left(x_k - x_j\right)\Big/2A_e\ , \tag{50}$$

where A_e is the area of the T3 element formed by nodes i, j, and k arranged in a counterclockwise order. Within the T3 element domain, the bubble function is characterized by the following properties:

$$N_u^b > 0, \forall \boldsymbol{x} \in \Omega_e; \quad N_u^b = 0, \forall \boldsymbol{x} \in \partial\Omega_e; \quad N_u^b(\boldsymbol{x}_{\text{centroid}}) = 1.0\ . \tag{51}$$

As shown in Fig. 3, the edge-based smoothing domain Ω_k is constructed by connecting the two endpoints of the shared edge with the centroids (bubble nodes) of the adjacent elements. In this domain, the smoothed strain $\bar{\boldsymbol{\varepsilon}}$ is expressed as follows:

$$\bar{\boldsymbol{\varepsilon}} = \frac{1}{A_k}\int_{\Omega_k} \overline{\boldsymbol{B}}_u \boldsymbol{u}\mathrm{d}\Omega = \sum_{\zeta\in\Psi^k} \overline{\boldsymbol{B}}_\zeta \boldsymbol{U}_\zeta\ , \tag{52}$$

in which ζ denotes the node of the nodal set Ψ^k within the smoothing domain. A_k represents the area of the smoothing domain, and $\boldsymbol{U}_\zeta$ denotes the unknown displacement at node ζ. By substituting Eqs. (48) and (49) into Eq.(52), the matrix formulation of the strain matrix $\overline{\boldsymbol{B}}_\zeta$ is derived as follows:

$$\begin{aligned} \boldsymbol{B}_\zeta &= \begin{bmatrix} b_{\zeta x}(\boldsymbol{x}_k) & 0 \\ 0 & b_{\zeta y}(\boldsymbol{x}_k) \\ b_{\zeta y}(\boldsymbol{x}_k) & b_{\zeta x}(\boldsymbol{x}_k) \end{bmatrix}, \\ b_{\zeta h}(\boldsymbol{x}_k) &= \frac{1}{A_k}\int_{\Gamma_k} N_\zeta n_h \mathrm{d}\Gamma = \frac{1}{A_k}\sum_{j=1}^{n_{eg}} N_\zeta(\boldsymbol{x}_{Gj}) n_{hj} L_j,\ (h = x, y)\ . \end{aligned} \tag{53}$$

In Eq. (53), n_{eg} denotes the number of line segments forming the smoothing domain, and L_j represents the length of the j th segment. $N_\zeta(\boldsymbol{x}_{Gj})$ is the shape function value at the Gaussian integration point G (indicated by the green marker in Fig. 3) of the j th segment, and n_{hj} denotes the component of the segment's unit outward normal vector.

3.3. SUPG method

In traditional FEM, the standard Galerkin formulation is typically employed to discretize problems governed by self-adjoint operators, yielding symmetric stiffness matrices that ensure uniqueness and stability of the solution. However, for problems involving convection, the convection operator is inherently non-self-adjoint. In convection-dominated scenarios, the standard Galerkin method yields a discretization equivalent to a central differencing scheme that assigns equal weights to both upstream and downstream nodes (Brooks and Hughes, 1982; Tezduyar and Osawa, 2000). This uniform weighting produces a stiffness matrix with significant antisymmetric components that capture the convective behavior, which in turn leads to spurious oscillations in the solution. To eliminate these spurious oscillations, the SUPG method is adopted in this study, wherein test functions are modified along the streamline to enhance the weighting of upstream contributions. In contrast to the conventional Galerkin approach where trial and test functions are identical, the SUPG formulation enhances its test function by adding a velocity-dependent stabilization term, as follows:

$$\overline{\boldsymbol{N}}_\theta = \boldsymbol{N}_\theta + \tau_s \boldsymbol{v}^T(\nabla \boldsymbol{N}_\theta)\ = \boldsymbol{N}_\theta + \tau \boldsymbol{v}^T \boldsymbol{B}_\theta\ , \tag{54}$$

where N_θ represents the shape function associated with the physical field θ. For the present transient convection–diffusion-reaction-type transport problem, the stabilization parameter τ is evaluated locally at the element level using the inverse-square-root formula recommended by Donea and Huerta (Donea and Huerta, 2003):

$$\tau = \left[\left(\frac{2}{\Delta t}\right)^2 + \left(\frac{2\|\boldsymbol{v}_c\|}{h}\right)^2 + \left(\frac{4\kappa}{h^2}\right)^2\right]^{-1/2}\ , \tag{55}$$

where Δt is the time increment, $\mathbf{v}_c$ denotes the local convection vector, and κ is the diffusivity. For linear triangular elements, the characteristic length is taken as $h = 2\sqrt{A_e}$, with A_e being the element area. This definition incorporates the transient, advective, and diffusive time scales in a unified manner and is consistent with element-level stabilization-parameter concepts based on local element information (Tezduyar and Osawa, 2000). The SUPG formulation provides stabilization mainly through streamline-oriented numerical dissipation, but it is not inherently monotonicity-preserving or discrete-maximum-principle-preserving. Localized over- or undershoots may therefore still occur near steep reaction fronts, which is particularly important for chemically reacting transport because small boundedness violations can be amplified by nonlinear reaction kinetics (Burman and Ern, 2002). Accordingly, SUPG is adopted here to mitigate convection-induced oscillations, rather than to guarantee complete removal of all front-related numerical artifacts. For sharper reaction fronts or stronger reaction sensitivity, supplementary discontinuity-capturing or nonlinear-diffusion stabilization may be required (Tezduyar and Park, 1986).

Applying the SUPG scheme allows for the derivation of the weak form of the energy conservation equation, which is presented below:

$$\begin{aligned} \mathscr{L}_T = &\int_\Omega \left(\rho c \overline{\boldsymbol{N}}_T^{\mathrm{T}} \dot{T} + \rho_g c_g \overline{\boldsymbol{N}}_T^{\mathrm{T}} \cdot \boldsymbol{q}_g \cdot \nabla T + \rho_w c_w \overline{\boldsymbol{N}}_T^{\mathrm{T}} \cdot \boldsymbol{q}_w \cdot \nabla T\right)\mathrm{d}\Omega \\ &+ \int_\Omega (k_T \nabla \boldsymbol{N}_T \cdot \nabla T)\mathrm{d}\Omega + \int_{\Gamma_N} \overline{\boldsymbol{N}}_T^{\mathrm{T}} Q_N \mathrm{d}\Gamma \\ &- \int_{\Gamma_R} \overline{\boldsymbol{N}}_T^{\mathrm{T}} h_r (T_r - T)\mathrm{d}\Gamma + \int_\Omega \overline{\boldsymbol{N}}_T^{\mathrm{T}} M_h R_h \Delta H \mathrm{d}\Omega \end{aligned}\ . \tag{56}$$

Accordingly, spatial discretization leads to the following matrix form of the energy conservation equation:

$$(\boldsymbol{K}_{TT} + \boldsymbol{K}_{TT}^{supg})\dot{\boldsymbol{T}} + (\boldsymbol{K}_T + \boldsymbol{K}_T^{supg})\boldsymbol{T} = \boldsymbol{F}^h + \boldsymbol{F}^{h-supg}\ , \tag{57}$$

with,

$$\begin{aligned} \boldsymbol{K}_{TT}^{supg} &= \textstyle\int_\Omega \left(\rho c \tau_s \boldsymbol{B}_T^T \boldsymbol{v} \boldsymbol{N}_T\right)\mathrm{d}\Omega\ , \\ \boldsymbol{K}_T^{supg} &= \int_\Omega \left(\rho_g c_g \tau_s \boldsymbol{B}_T^T \boldsymbol{v} \boldsymbol{q}_g^T \boldsymbol{B}_T + \rho_w c_w \tau_s \boldsymbol{B}_T^T \boldsymbol{v} \boldsymbol{q}_w^T \boldsymbol{B}_T\right)\mathrm{d}\Omega\ , \end{aligned} \tag{58}$$

$$\begin{aligned} \boldsymbol{F}^{h-supg} = &-\int_{\Gamma_N} \boldsymbol{B}_T^T \boldsymbol{v} \tau_s Q_N \mathrm{d}\Gamma + \int_{\Gamma_R} \boldsymbol{B}_T^T \boldsymbol{v} \tau_s h_r (T_r - T)\mathrm{d}\Gamma \\ &- \int_\Omega \boldsymbol{B}_T^T \boldsymbol{v} \tau_s M_h R_h \Delta H \mathrm{d}\Omega\ . \end{aligned} \tag{59}$$

Similarly, solving the gas–water mixture mass conservation equation under convection-dominated conditions often leads to spatial oscillations, which motivates the use of the SUPG scheme. Thus, a simple two-phase flow problem is presented to illustrate the detailed SUPG discretization process. Neglecting the effects of other phases and physical fields, the strong form of mass conservation equation in the two-phase flow model derived from the current multi-physics framework is as follows:

$$\frac{\partial s_w}{\partial t} + \boldsymbol{q}_t \cdot \nabla\left(\frac{s_w^2}{s_w^2 + s_g^2}\right) = 0\ , \tag{60}$$

in which $\boldsymbol{q}_t$ represents the total flux, and the term within the gradient operator corresponds to the fractional flow function defined by the phase saturations. A detailed derivation of Eq. (60) is given in Appendix-C. Using the SUPG method, its weak form is derived as follows:

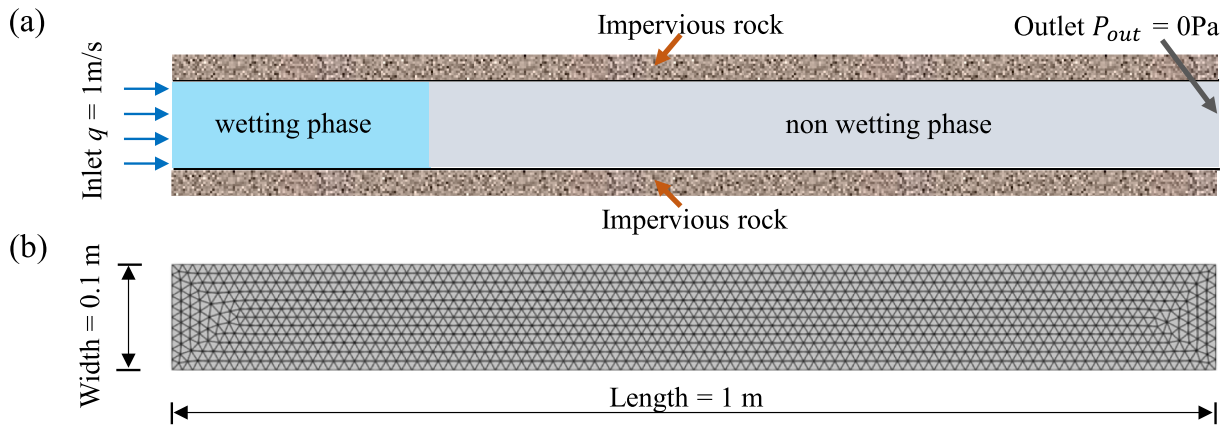

Fig. 4. Two-Phase Flow Model: (a) schematic with imposed boundary conditions and (b) mesh structure.

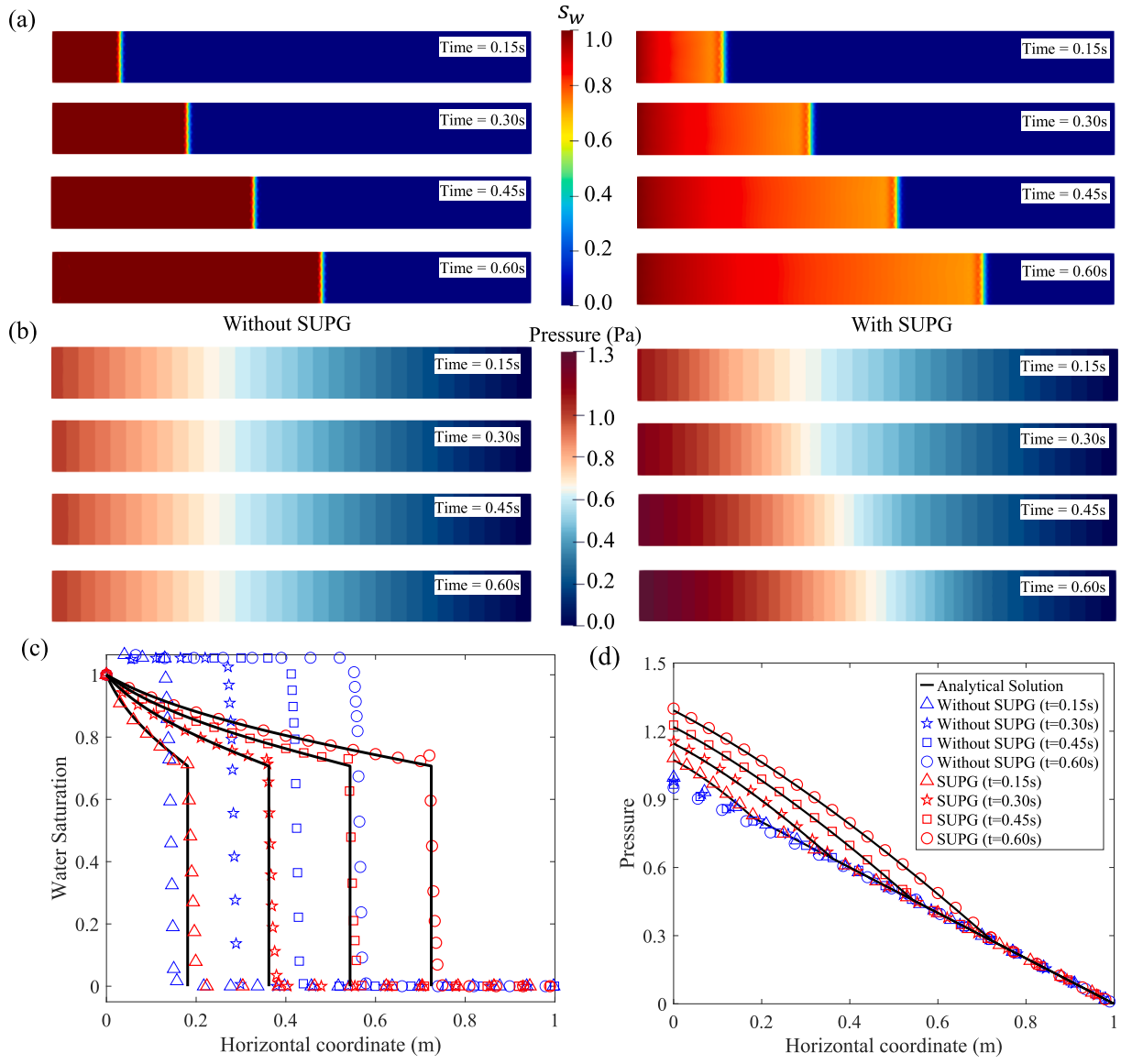

Fig. 5. Comparison of numerical results with and without the SUPG scheme; (a) spatial distributions of water saturation at various time instants, (b) pressure contours at various time instants, (c) water saturation and (d) pressure distribution along the flow path at different times.

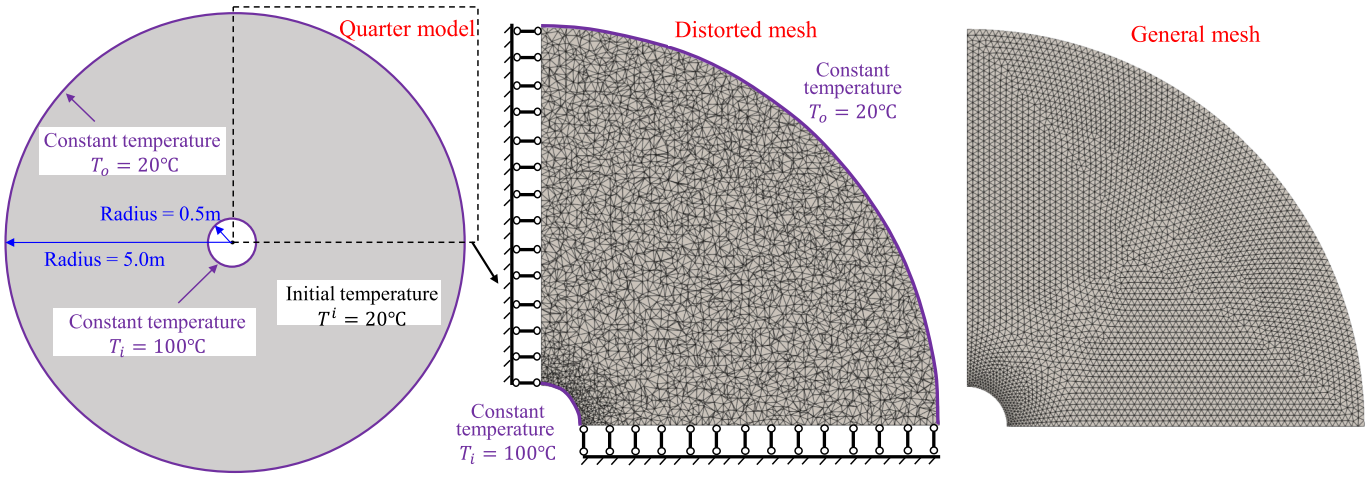

Fig. 6. Distorted and general meshes of the hollow disc model with boundary conditions.

Table 1
Material parameters in hollow disc model.

Parameter	Value	Parameter	Value
Elastic modulus	$E = 5$GPa	Poisson's ratio	$\nu = 0.25$
Density	$\rho_s = 2500$ k g/m^3	Specific heat	$c_s = 1000$ J/(kg·K)
Thermal conductivity	k_{Ts}=1W/(m·K)	linear thermal expansion coefficient	$\alpha_s = 5.0 \times 10^{-6}$/K

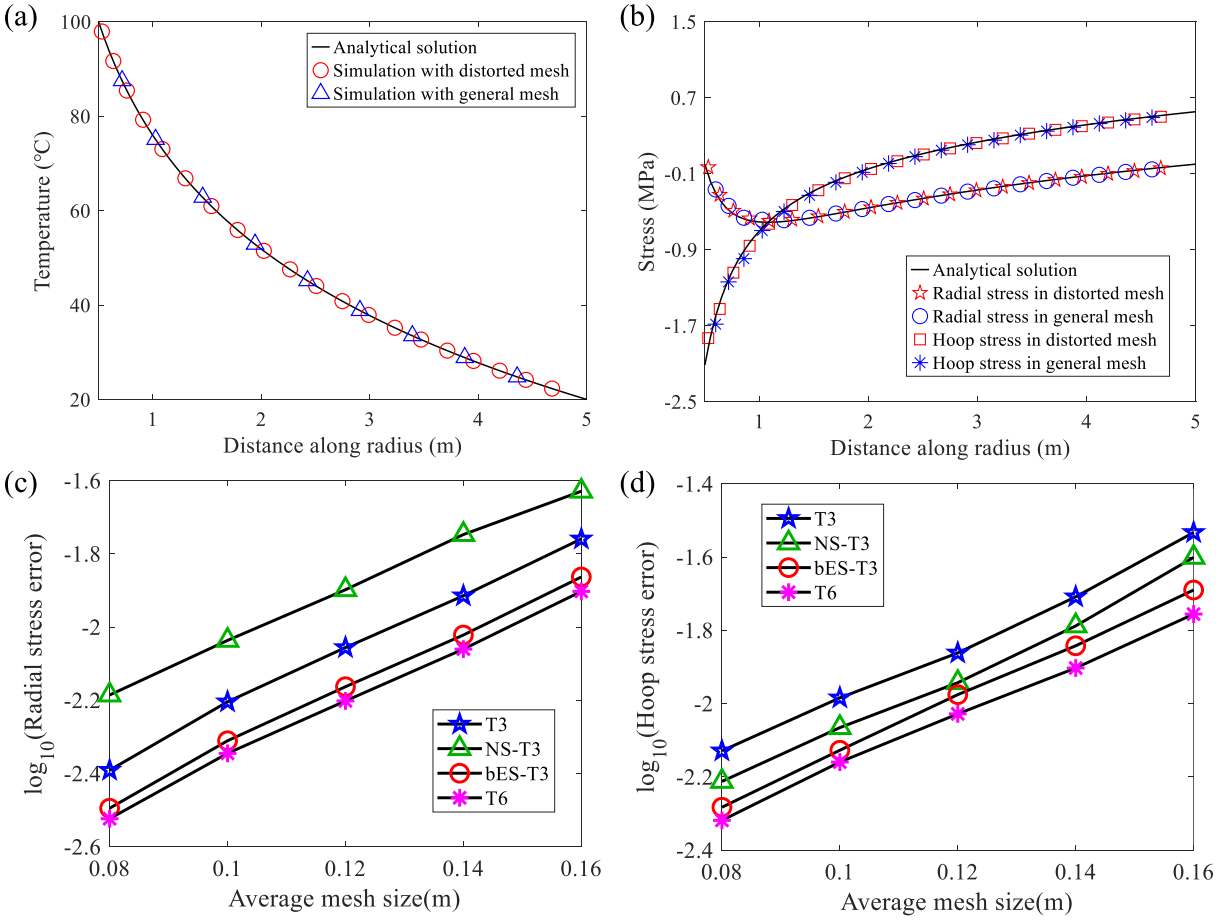

Fig. 7. Comparison between numerical and analytical solutions: (a) temperature evolution along the radial direction; (b) radial and hoop stress evolution along the radial direction; convergence of the error norms for (c) radial stress and (d) hoop stress with respect to the average mesh size.

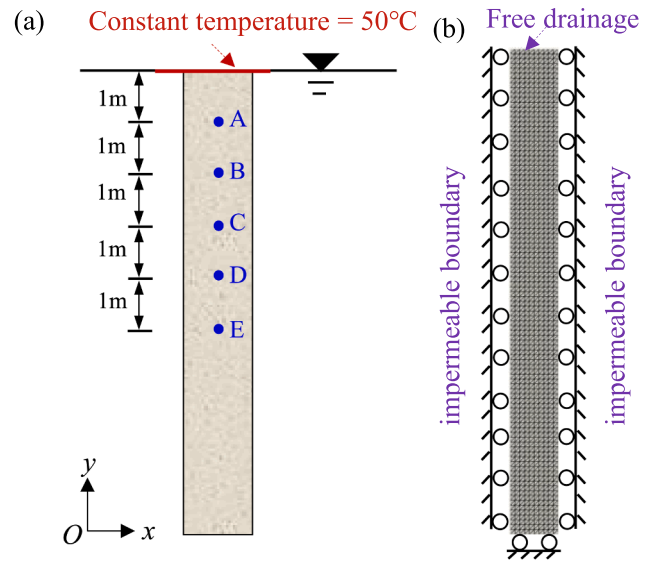

Fig. 8. (a) Geometric model with boundary conditions and (b) corresponding mesh.

Table 2
Material parameters in the semi-infinite domain.

Parameter	Value
Porosity	$\phi = 0.2$
Elastic modulus	$E = 6$ MPa
Poisson's ratio	$\nu = 0.4$
Mixture density	$\rho = 1000$ kg/m^3
Specific heat capacity	$c = 167.2$ J/(kg·℃)
Permeability coefficient	k = 1.556 × 10^{-5} m/s
Thermal conductivity	$k_T = 836$ W/(m·℃)
Thermal expansion coefficient	$\alpha = 3.0 \times 10^{-5}$ 1/K

$$\begin{aligned}\mathscr{L}_T &= \int_\Omega \left(\overline{\boldsymbol{N}}^T \dot{s}_w + \overline{\boldsymbol{N}}^T \boldsymbol{q}_t \cdot \nabla \left(\frac{s_w^2}{s_w^2 + s_g^2} \right) \right) \mathrm{d}\Omega \\ &= \int_\Omega (\boldsymbol{N}^T + \tau_s \nabla \boldsymbol{N}^T \boldsymbol{q}_t) \dot{s}_w \mathrm{d}\Omega \\ &+ \int_\Omega (\boldsymbol{N}^T + \tau_s \nabla \boldsymbol{N}^T \boldsymbol{q}_t) \boldsymbol{q}_t \cdot \nabla \left(\frac{s_w^2}{s_w^2 + s_g^2} \right) \mathrm{d}\Omega \,,\end{aligned} \tag{61}$$

Based on the weak form in Eq. (61), the water saturation is updated explicitly, and the pressure is subsequently obtained by implicitly solving the hybrid mass conservation equation for the gas–water mixture to suppress spatial oscillations.

4. Verifications

To guarantee the proposed unified framework serves as a reliable computational tool for multi-physics coupling analysis, this section presents a systematically structured series of benchmark validations. These tests are specifically designed to verify the framework's accuracy and stability across progressively complex multi-physics scenarios. The sequence commences with a two-phase flow simulation, which confirms the accuracy of capturing immiscible fluid displacement and demonstrates the SUPG technique's ability to resolve convection-dominated transport without numerical oscillation. Subsequently, the framework's performance in handling solid deformation coupled with thermal and hydraulic processes is evaluated. A thermo-mechanical (TM) heat conduction problem first highlights the high accuracy and low mesh sensitivity inherent to the bES-FEM formulation. The complexity is then elevated to a thermo-hydro-mechanical (THM) heating simulation, verifying the method's robustness in managing intricate multi-physics interactions under challenging low-permeability conditions. Finally, the framework's comprehensive capability is validated against the Masuda's hydrate dissociation test. This fully coupled THMC simulation proves the tool's overall effectiveness in modeling the highly nonlinear phase transitions and multiphase flow behaviors characteristic of methane hydrate dissociation.

4.1. Two-phase flow model

Within unsaturated porous media, the coexistence of distinct immiscible fluid phases typically leads to complex multiphase flow behavior. To evaluate the current computational framework for multiphase flow, the classical two-phase flow model (also known as Buckley-Leverett problem (Buckley and Leverett, 1942) was selected as a benchmark. In this model, the transient process of a wetting phase displacing a non-wetting phase in a horizontal porous medium is simulated, neglecting gravity and capillary effects. Standard FEM is prone to spurious oscillations in advection-dominated flows, requiring stabilization remedies like SUPG or artificial diffusion terms. Since the present framework integrates the SUPG scheme, this two-phase flow model serves as an ideal benchmark to validate its numerical performance particularly given the problem's purely advective nature.

Fig. 4 illustrates the geometry and the computational mesh, which consists of 1304 nodes and 2386 elements. The model is initially dominated by the non-wetting phase ($s_{nw} = 0.9999, s_w = 0.0001$). A constant water injection velocity of 1.0 m/s is applied at the inlet with a water saturation of $s_w = 1.0$, while a constant pressure of zero is maintained at the outlet. The fluid flow follows Darcy's law, where the velocity is assumed to be a quadratic function of saturation ($v \propto s^2$), and other physical parameters are set to unity for simplicity. The analytical solution for this classical problem is given by:

$$\begin{aligned}\frac{x}{t} &= \begin{cases} \dfrac{1+\sqrt{2}}{2}, & 0 \leqslant s_w \leqslant \dfrac{\sqrt{2}}{2} \\ \dfrac{2s_w(1-s_w)}{\left(2s_w^2 - 2s_w + 1\right)^2}, & \dfrac{\sqrt{2}}{2} < s_w \leqslant 1 \end{cases}, \\ p(x) &= p(1) - \int_x^1 \frac{dp}{dx}\,\mathrm{d}x = \int_x^1 \frac{1}{\left(2s_w^2 - 2s_w + 1\right)}\,\mathrm{d}x \,.\end{aligned} \tag{62}$$

Fig. 5(a) presents a comparison of water saturation distributions computed with the SUPG method against those obtained without it at different time points, while Fig. 5(b) contrasts the corresponding pressure contours between the two approaches. Fig. 5(c) compares the water saturation distribution along the flow direction at various time points with the analytical solution. Without the SUPG technique, the shock front exhibits severe spurious oscillations, leading to marked differences between the numerical and analytical solutions. Conversely, applying the SUPG technique results in a more robust water saturation profile; the numerical results closely match the analytical solution and precisely capture the saturation front. Furthermore, Fig. 5(d) demonstrates that simulations without SUPG produce unrealistic pressure distributions near the shock front, while the incorporation of SUPG yields more stable and accurate pressure profiles. These findings confirm that the SUPG-enhanced framework exhibits superior stability and accuracy in convection-dominated scenarios. Consequently, the hydraulic module within the developed THMC coupling framework can accurately capture the multiphase flow behavior, particularly exhibiting robust stability and reliability under convection-dominated conditions.

4.2. TM coupling: Heat conduction in hollow disc

We consider heat conduction in a hollow disc with an internal radius $r_i = 0.5$m and external radius $r_o = 5$m. Due to the geometric symmetry and plane strain conditions, only a quarter of the geometry is modelled with symmetric boundaries, as shown in Fig. 6. The model is initialized

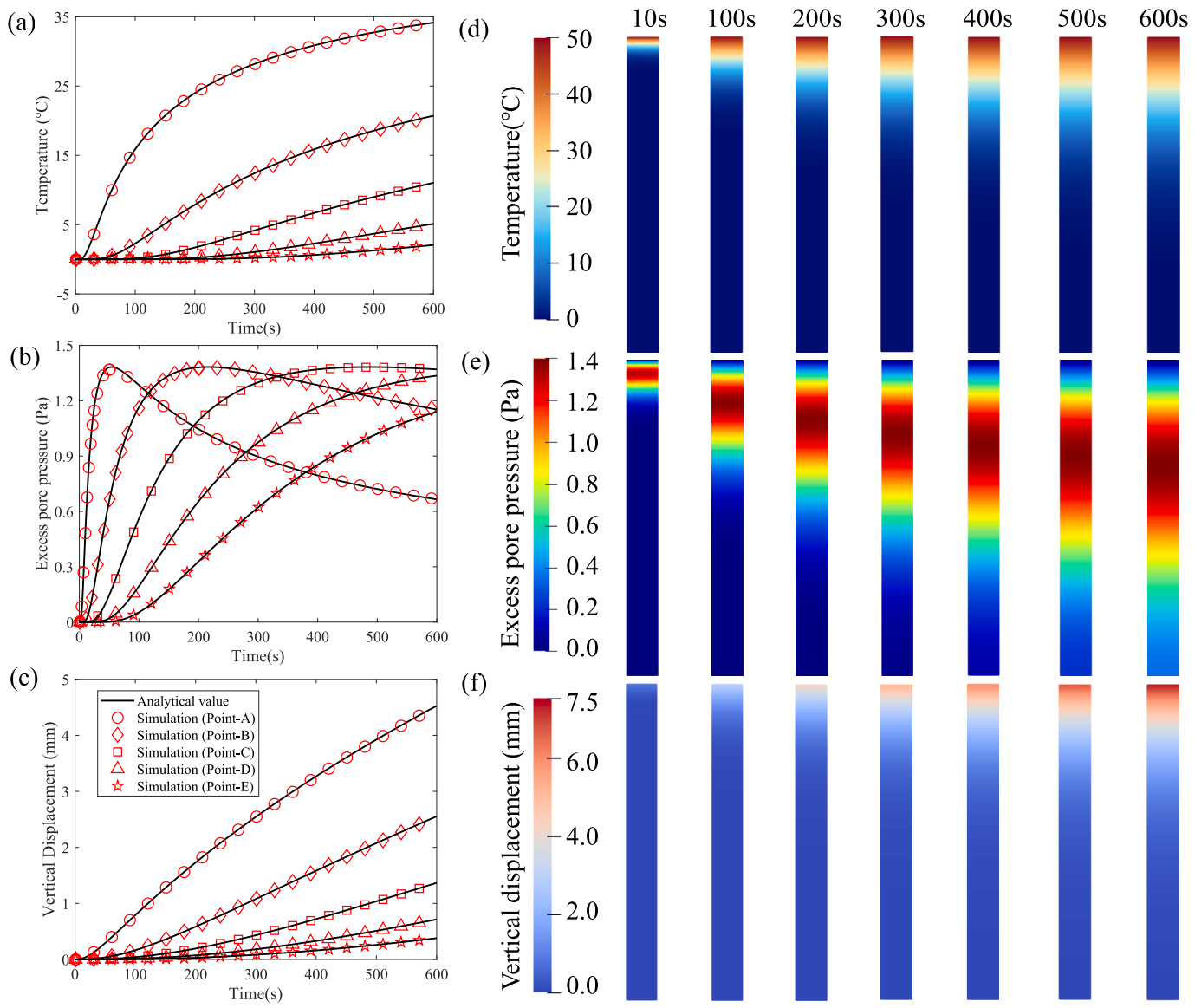

Fig. 9. Heating model in a semi-infinite domain: (a-c) numerical versus analytical comparisons for vertical temperature, excess pore pressure, and vertical displacement at five designated points; (d-f) Contours of temperature, excess pore pressure, and vertical displacement at different time instants.

at 20°C, and its boundary conditions are also illustrated in Fig. 6. The material parameters of the model are presented in Table 1. To demonstrate the capability of the current framework in simulating the TM coupling problems and overcoming mesh distortion, two mesh configurations are employed: a distorted mesh and a regular mesh. Both mesh types contain 3903 nodes and 7557 elements. Upon reaching thermal equilibrium, the temperature, radial stress, and hoop stress satisfy the following analytical expressions:

$$\begin{cases} T(r) = T_i + \dfrac{T_o - T_i}{\ln(r_o/r_i)}\ln(r/r_i) \\ \sigma_r(r) = \dfrac{E\alpha(T_i - T_o)}{2(1-\nu)\ln(r_o/r_i)}\left[\ln\left(\dfrac{r}{r_i}\right) - \dfrac{r_o^2}{r_o^2 - r_i^2}\left(1 - \dfrac{r_i^2}{r^2}\right)\ln\left(\dfrac{r_o}{r_i}\right)\right] \\ \sigma_h(r) = \dfrac{E\alpha(T_i - T_o)}{2(1-\nu)\ln(r_o/r)}\left[1.0 + \ln\left(\dfrac{r}{r_i}\right) - \dfrac{r_o^2}{r_o^2 - r_i^2}\left(1 + \dfrac{r_i^2}{r^2}\right)\ln\left(\dfrac{r_o}{r_i}\right)\right] \end{cases} \tag{63}$$

Fig. 7 (a)~(b) illustrates the radial distribution curves of the simulated temperature and stress fields. The results from both the regular and distorted meshes closely match the analytical solution, demonstrating that the current bES-FEM method exhibits robustness against mesh distortion and high accuracy in addressing TM coupling problems. To further validate the accuracy of the proposed method, numerical results from NS-FEM, T3-FEM, and T6-FEM were compared with those from bES-FEM. Fig. 7 (c)~(d) presents the numerical errors of the simulated radial and hoop stresses against the analytical solutions for various average mesh sizes. It is observed that the numerical error increases with mesh size for all four methods. For a given average mesh size, both NS-FEM and T3-FEM yield larger errors, while T6-FEM exhibits the smallest numerical error. Although T6-FEM achieves higher accuracy than bES-FEM, it also entails more degrees of freedom and increased computational cost. Therefore, the current bES-FEM can be regarded as an effective method for balancing computational accuracy and efficiency.

4.3. THM coupling: Heating of semi-infinite domain

To assess the effectiveness of the current framework in simulating THM coupling problems, the classic problem of a saturated soil layer subjected to heating in a semi-infinite domain (Bai and Abousleiman,

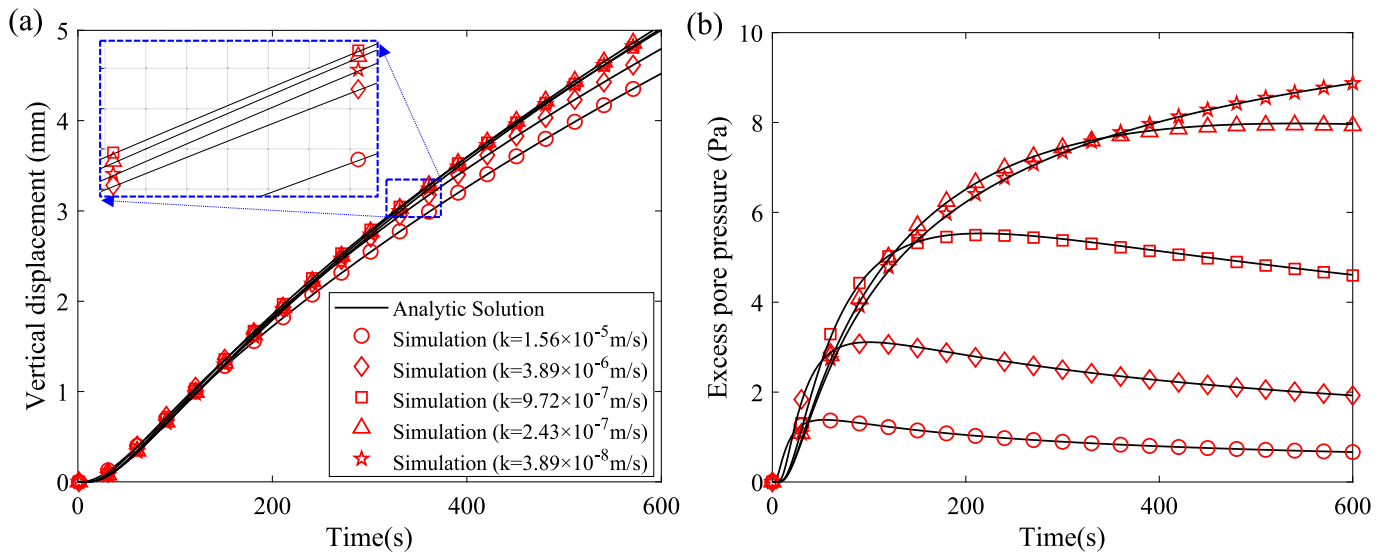

Fig. 10. Comparative Analysis of numerical and analytical temporal evolution curves for (a) vertical displacement and (b) excess pore pressure at point-A under different permeability coefficients.

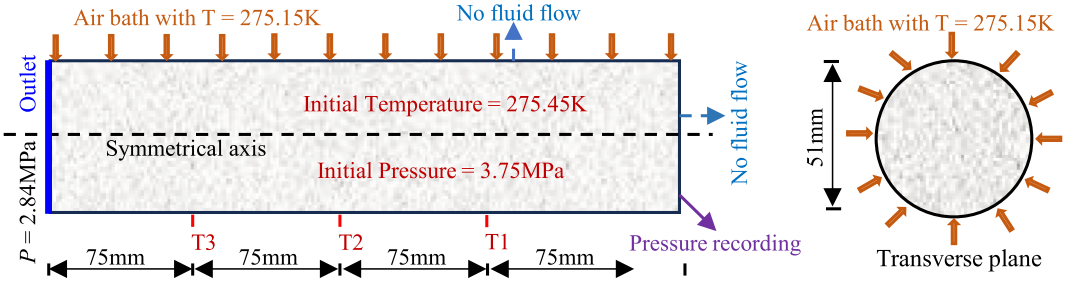

Fig. 11. Schematic of the geometry and boundary conditions for the Masuda's model.

Table 3
Material parameters for Masuda's model (Tang et al., 2026).

Parameters	Value	Parameters	Value	Parameters	Value
ρ_s	2675 kg/m^3	k_{Ts}	2.9 W/(m • K)	α_s	1.8×10^{-6}K^{-1}
ρ_w	1000 kg/m^3	k_{Th}	0.46 W/(m • K)	M_h	0.124 kg/mol
ρ_h	910 kg/m^3	k_{Tw}	0.56 W/(m • K)	M_g	0.016 kg/mol
c_s	750J/(kg • K)	k_{Tg}	0.07 W/(m • K)	M_w	0.018 kg/mol
c_w	4200J/(kg • K)	R	8.3145	s_{wr}	0.1
c_h	2200J/(kg • K)	$\Delta E/R$	−9752.73 K	s_{gr}	0.02
c_g	2180J/(kg • K)	k_{d0}	36000mol/(m^2 • Pa • s)	N	10
μ_w	10^{-3} kg/(m^2 • s)	k_{h0}	9.798×10^{-14} m^2		

1997; Hua et al., 2026) is tested. The geometric model, boundary conditions, and mesh configuration are presented in Fig. 8; the mesh comprises 2000 elements and 1111 nodes. Neglecting phase change processes, the model is initialized at 0°C with a constant temperature boundary of 50°C applied at the top surface. The model parameters are detailed in Table 2. An exact solution for this well-known model is available in Ref (McTigue, 1986). To facilitate comparative analysis, five points (labeled A, B, C, D, and E) are measured at 1 m intervals over a 5 m range from the top surface.

Fig. 9 (a)~(c) compare the numerical and analytical temporal evolution curves of temperature, excess pore water pressure and vertical displacement at five observation points, illustrating excellent agreement and confirming the method's accuracy for THM coupling. Fig. 9 (d)~(f) present contours of the temperature, excess pore pressure and vertical displacement fields at various time instants. It is evident that the constant high temperature imposed on the top surface promotes the downward heat conduction, leading to a progressive temperature increase in the underlying saturated soil. This, in turn, induces thermal expansion that results in a positive vertical displacement. Meanwhile, the high temperature at the upper boundary induces volumetric deformation in the surrounding soil skeleton, resulting in an initial increase in excess pore pressure. As consolidation progresses and temperature evolves, the peak excess pore pressure gradually shifts downward while the initial pore pressure diminishes. To demonstrate the feasibility of the current approach under nearly undrained conditions with low permeability coefficients, four supplementary cases with reduced permeability (i.e., $k = 3.89\times10^{-6}$, 9.72×10^{-7}, 2.43×10^{-7}, and 3.89×10^{-8}) were simulated while keeping the other parameters constant. Fig. 10 compares the numerical and analytical solutions at point-A for five distinct permeability coefficients, showing the temporal curves for vertical displacement and excess pore pressure. The results indicate that lowering the permeability does not weaken the consistency between

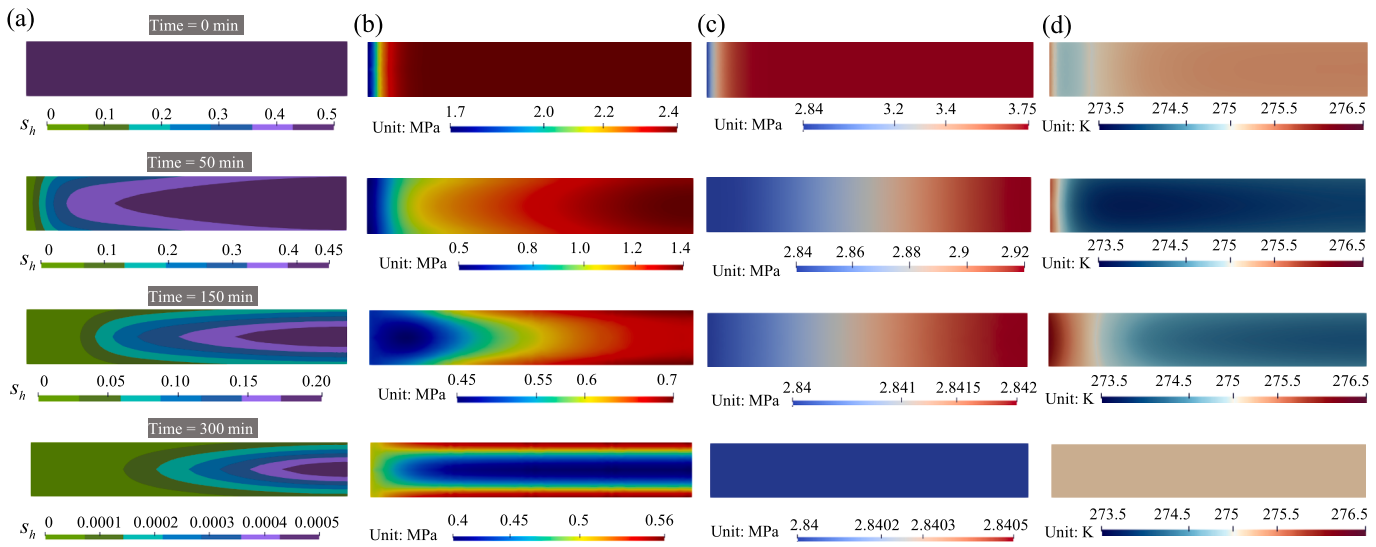

Fig. 12. Distributions of (a) hydrate saturation, (b) axial stress, (c) pressure, and (d) temperature over different time intervals.

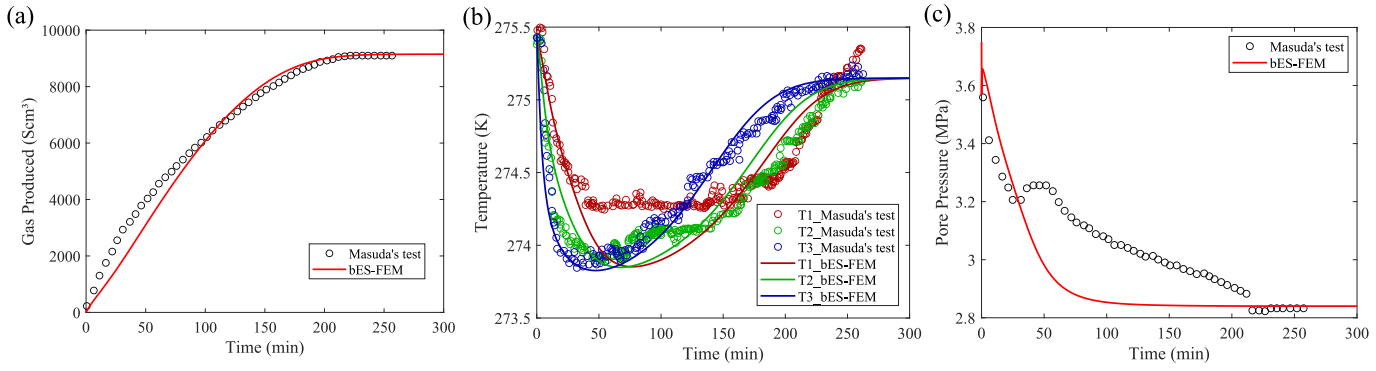

Fig. 13. Comparative analysis of numerical results and Masuda's experimental measurements.

numerical and analytical solutions, highlighting the robustness of the proposed framework when handling nearly incompressible conditions via bES-FEM.

4.4. THMC coupling: Masuda's hydrate dissociation model

To validate the effectiveness of the current framework in addressing THMC coupling problems, an axisymmetric simulation of Masuda's hydrate dissociation model (Masuda, 1999) is conducted. Fig. 11 presents the geometry and boundary conditions of Masuda's cylindrical model. At the initial state, the porous medium exhibits a porosity of 0.182. The corresponding saturations are 0.501 for methane hydrate, 0.351 for water, and 0.148 for gas. A constant pressure boundary condition of 2.84 MPa is imposed at the left outlet, whereas all other boundaries are impermeable to fluid flow. The circumferential side is maintained in a constant-temperature air bath. The material parameters used in the Masuda benchmark are summarized in Table 3. They follow the numerical interpretation of Masuda's experiment in Ref (Tang et al., 2026), where these parameters have been examined through benchmark validation and sensitivity analyses. Accordingly, these key parameters such as the permeability reduction exponent N can be considered to ensure reliable reproduction of the dissociation responses and a reasonable description of the permeability model. To monitor the temperature evolution during hydrate dissociation, average temperatures are recorded at three cross-sectional planes (T1, T2, and T3) positioned 75 mm apart within the model.

Fig. 12 presents contours of various physical quantities at different time instants during the hydrate dissociation process. In the early stages, methane hydrate at the outlet undergoes endothermic dissociation, resulting in an immediate drop in local temperature. Although the local temperature drop might inhibit the endothermic process, the rapid decline in hydrate saturation at the outlet simultaneously promotes further dissociation. As hydrate dissociation proceeds, the pore pressure along the fluid flow gradually decreases, while the weakening confining effect in low-hydrate saturation regions results in reduced axial stress. At roughly 50 min, the endothermic dissociation of hydrates drives the model's temperature to its minimum, while sustained heating from the air bath further accelerates dissociation near the circumferential boundary. During the later stage of hydrate dissociation, temperature dominates the process, resulting in lower hydrate saturation near the circumferential boundary than in regions closer to the central axis. Moreover, after the rapid cooling induced by endothermic dissociation, the model undergoes accelerated rewarming and expansion from heat flux at the circumferential boundary, resulting in slightly higher axial stress compared to that in the core region.

Fig. 13 compares the current numerical results with the experimental data. The simulated gas production curve agrees well with experimental measurements. Although the temperature evolution at the T1 cross-section exhibits slight deviations from experimental data, the overall trend remains consistent. Notably, the evolution of the far-field pore pressure diverges significantly from the experimental observations after 30 min. This discrepancy may be attributed to the limited availability of actual experimental parameters and the simplified nature of the model representation. Fundamentally, the pressure evolution is influenced by

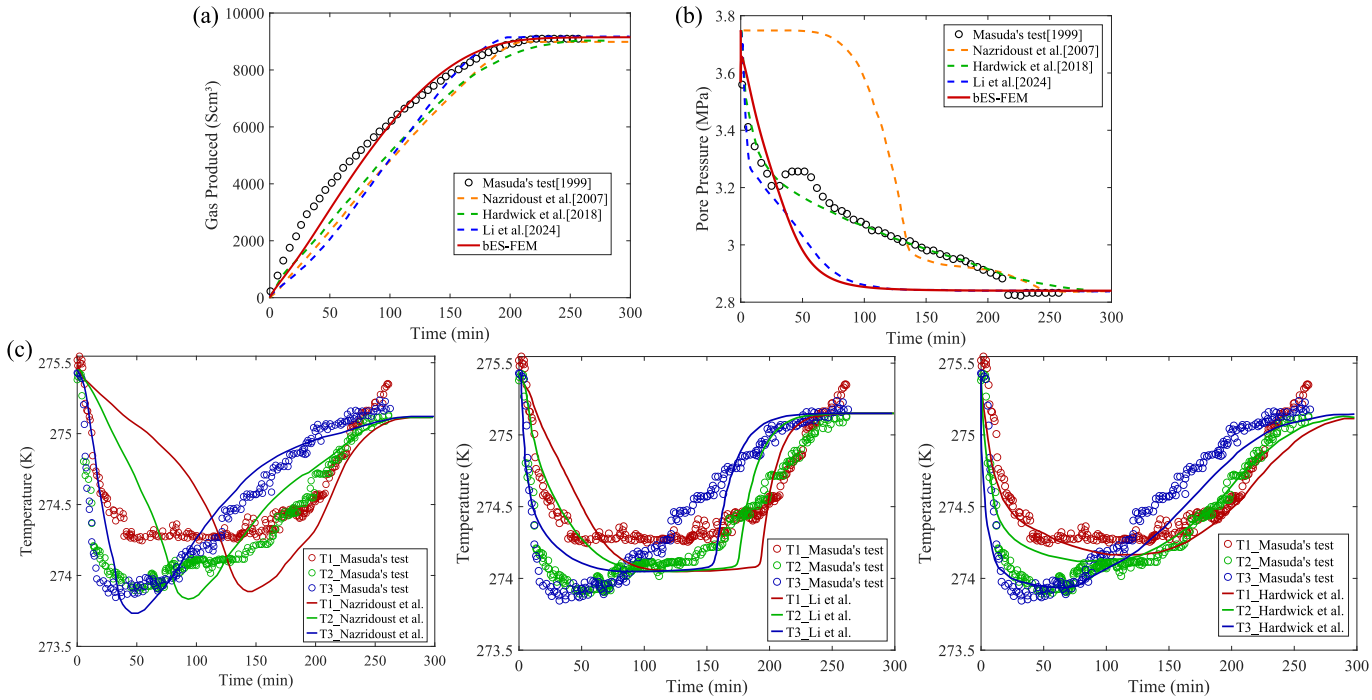

Fig. 14. Comparative analysis of various numerical results (Li et al., 2024; Hardwick and Mathias, 2018; Nazridoust and Ahmadi, 2007) with Masuda's experimental data.

multiple factors, including the hydrate dissociation rate model and the permeability model. Although many researchers have attempted to employ more advanced permeability models (Sun et al., 2005; Deng et al., 2020; Hardwick and Mathias, 2018; Song et al., 2021) and different permeability reduction coefficients (Ruan et al., 2012; Nazridoust and Ahmadi, 2007; Zhao et al., 2012) to achieve more accurate pressure evolution, these approaches have not been able to simultaneously achieve high accuracy in both gas production and temperature predictions. Fig. 14 compares the simulation results from other studies with experimental data. It is evident that most simulations show relatively poor agreement in predicting far-field pressure evolution. As indicated in Ref (Li et al., 2024), accurately capturing pressure evolution is highly challenging. Although most models reproduce similar gas production rates and temperature trends consistent with experiments, notable discrepancies remain at certain stages, highlighting challenges in fully capturing the system's behavior. In contrast, the current simulation shows the highest accuracy in reproducing both gas production and temperature evolution, indicating that the hydrate dissociation process is well captured and confirming the efficacy of the framework for THMC coupling problems within an acceptable range.

5. Applications to methane hydrate exploitation

Methane hydrate deposits are typically located hundreds to thousands of meters below the seafloor (see Fig. 15(a)), and their large-scale exploitation in deep-sea environments poses significant engineering challenges. Understanding the complex multi-physics coupling mechanisms during such production processes is therefore of critical importance. Motivated by this need, this section extends the unified framework to simulate large-scale methane hydrate production.

Following the refinement of the governing mathematical model, a three-layer stratigraphic configuration with distinct hydrate saturation levels is first constructed, as shown in Fig. 15(b). The hydrate reservoir is modeled using an axisymmetric geometry; the corresponding mesh and boundary conditions are depicted in Fig. 15(c), which comprises 4580 nodes and 8924 elements. The material parameters are identical to those used in Masuda's benchmark case presented in Section 4.4. To account for the deep-sea environment, the upper boundary of the model is subjected to the combined effects of overburden soil weight and hydrostatic pressure. Accordingly, the simulation is conducted in two sequential steps. First, a steady-state geostatic equilibrium is established throughout the sedimentary sequence under self-weight loading. Once this initial state is achieved, hydrate production is initiated by applying depressurization or heating at the production well. The upper boundary is 360 m below sea level, with a total vertical stress of 6 MPa imposed. Following the recommendation of Ref (Zhang et al., 2024), the initial temperature field is assumed to increase linearly with depth z according to $T = T_0 + 0.0433z$, resulting in a fixed temperature of 275.45 K at the bottom boundary. The steady-state distributions under self-weight prior to production are shown in Fig. 16. As expected, pore pressure, effective vertical stress, and temperature all exhibit linear distributions with depth, showing full agreement with analytical solutions. For production simulations, a constant pressure of 2.9 MPa and temperature of 275 K are prescribed at the left-side outlet to promote hydrate dissociation. The pressure and temperature along the top and bottom boundaries are maintained as Dirichlet conditions, consistent with the pre-production equilibrium state. Horizontal displacements are restrained on the left and right boundaries, while vertical displacement is fixed at the bottom.

Fig. 17 illustrates the dynamic response mechanisms of the reservoir during deep-sea methane hydrate production. The process initiates with depressurization at the exploitation well, rapidly forming a low-pressure funnel zone that propagates radially into the reservoir. This swift reduction in pore water pressure disrupts the original thermodynamic equilibrium, acting as the primary trigger for subsequent hydrate dissociation. Driven by this perturbation, the hydrates undergo significant phase transformation: the high-hydrate-saturation regions progressively shrink as the solid phase decomposes into flowing fluid phases (gas and water). Concurrently, the process induces substantial endothermic reactions, manifesting as a low-temperature zone originating at the exploitation well and progressively extending into the overburden and underburden. This "cold front" effect forms a significant negative feedback mechanism, whereby excessive temperature drops could inhibit further dissociation or promote secondary hydrate formation, thereby risking channel blockage. With constant temperature and low

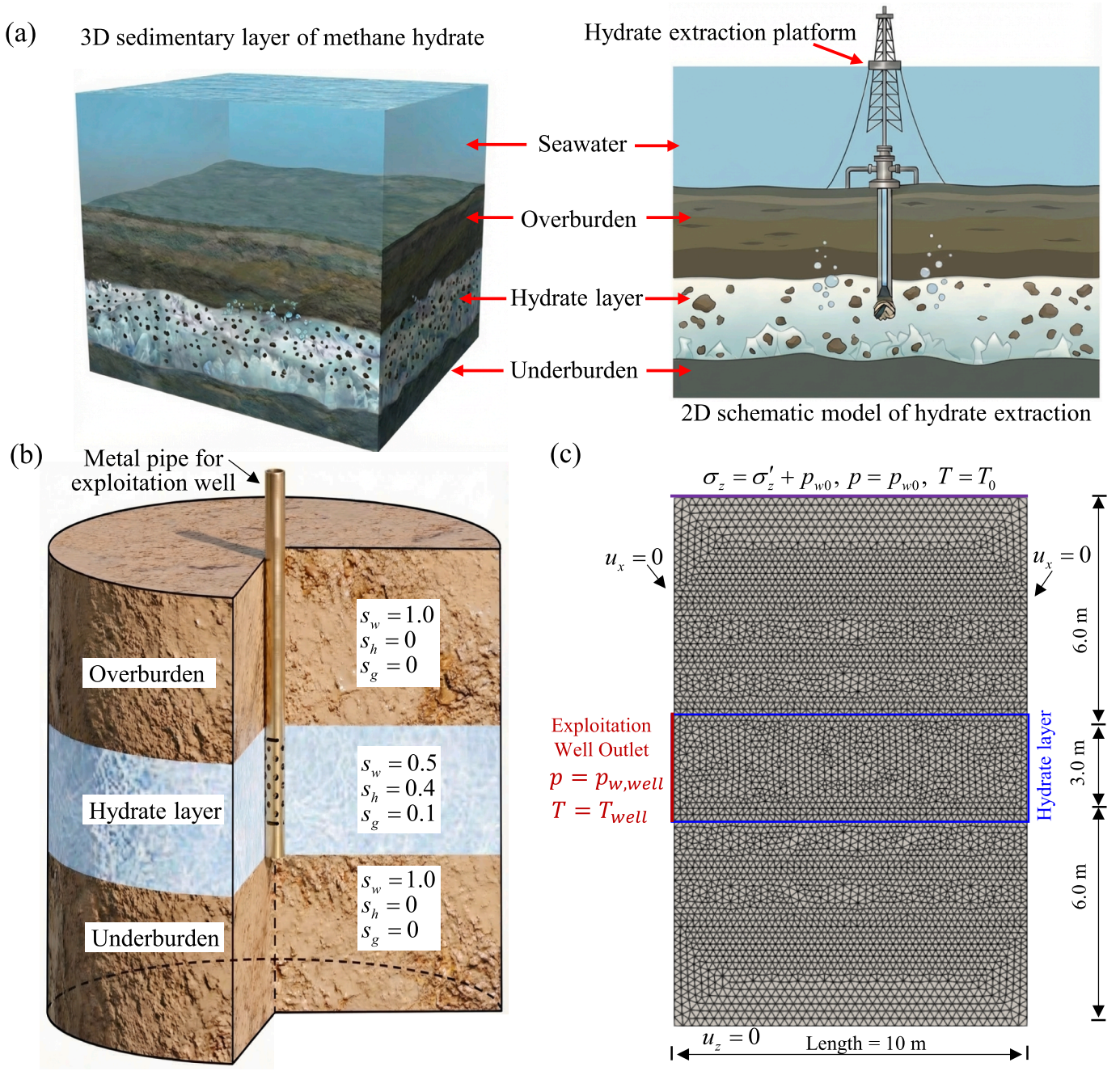

Fig. 15. (a) Schematic of a methane hydrate reservoir layer; (b) axisymmetric single-well production model; (c) discretized mesh and applied boundary conditions.

pressure maintained at the outlet, the early-stage production establishes a relatively stable pressure and temperature regime near the exploitation well, resulting in a consistent and uniform decline in nearby hydrate saturation. However, as production progresses, pressure and temperature in the radial direction away from the well enter a state of complex dynamic interplay. In this region, hydrate dissociation is jointly governed by temperature and pressure: low pressure promotes decomposition, while the endothermic cooling from dissociation inhibits further breakdown. This competing mechanism results in intermittent rather than continuous dissociation, manifesting as discontinuous and nonuniform reductions in hydrate saturation along the radial distance during the later stages. Furthermore, the hydraulic and thermal processes triggered by hydrate dissociation ultimately induce macroscopic geomechanical deformation. As evident from the vertical displacement contours, pore pressure depletion increases effective stress, compounded by the weakening of the sediment skeleton due to hydrate dissociation, thereby causing reservoir consolidation and induced subsidence in the overburden. Although depressurization exploitation efficiently promotes hydrate dissociation, the associated reservoir cooling effects and subsidence risks warrant careful consideration. Engineering strategies should incorporate targeted reinforcement at the reservoir-overburden/underburden interfaces in the vicinity of the production well to mitigate risks associated with large-deformation-induced failures. Informed by the present simulations, the integration of auxiliary heat injection is further suggested to counteract the risk of freeze-induced blockages near the exploitation well, complemented by the development of a robust geomechanical surveillance system aimed at optimizing gas recovery while safeguarding subsea stability.

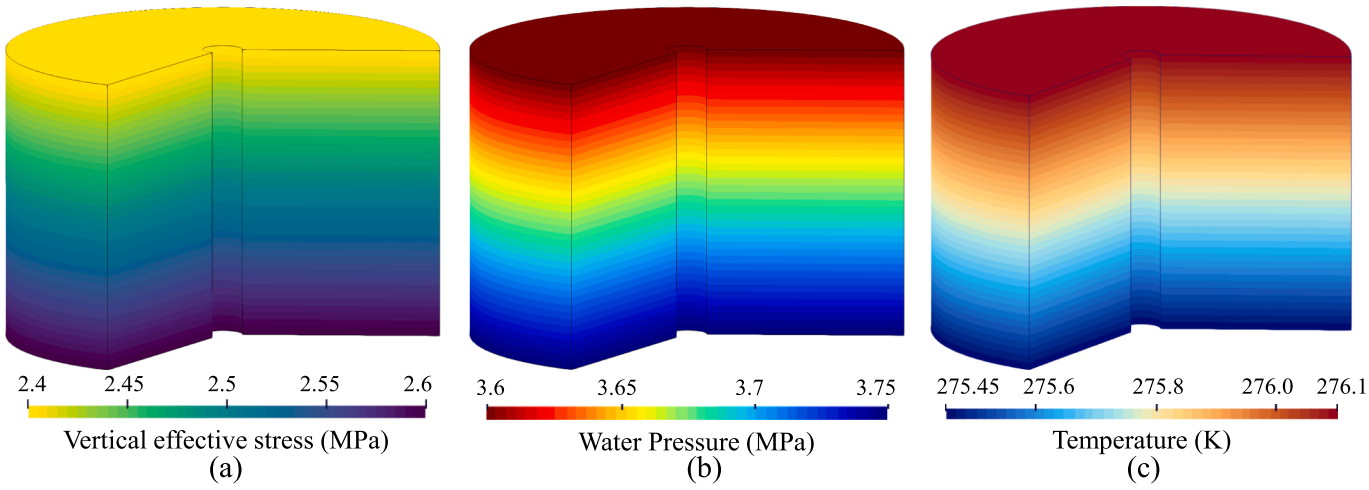

Fig. 16. Distributions of (a) vertical effective stress, (b) water pressure and (c) temperature under self-weight at steady state prior to hydrate production.

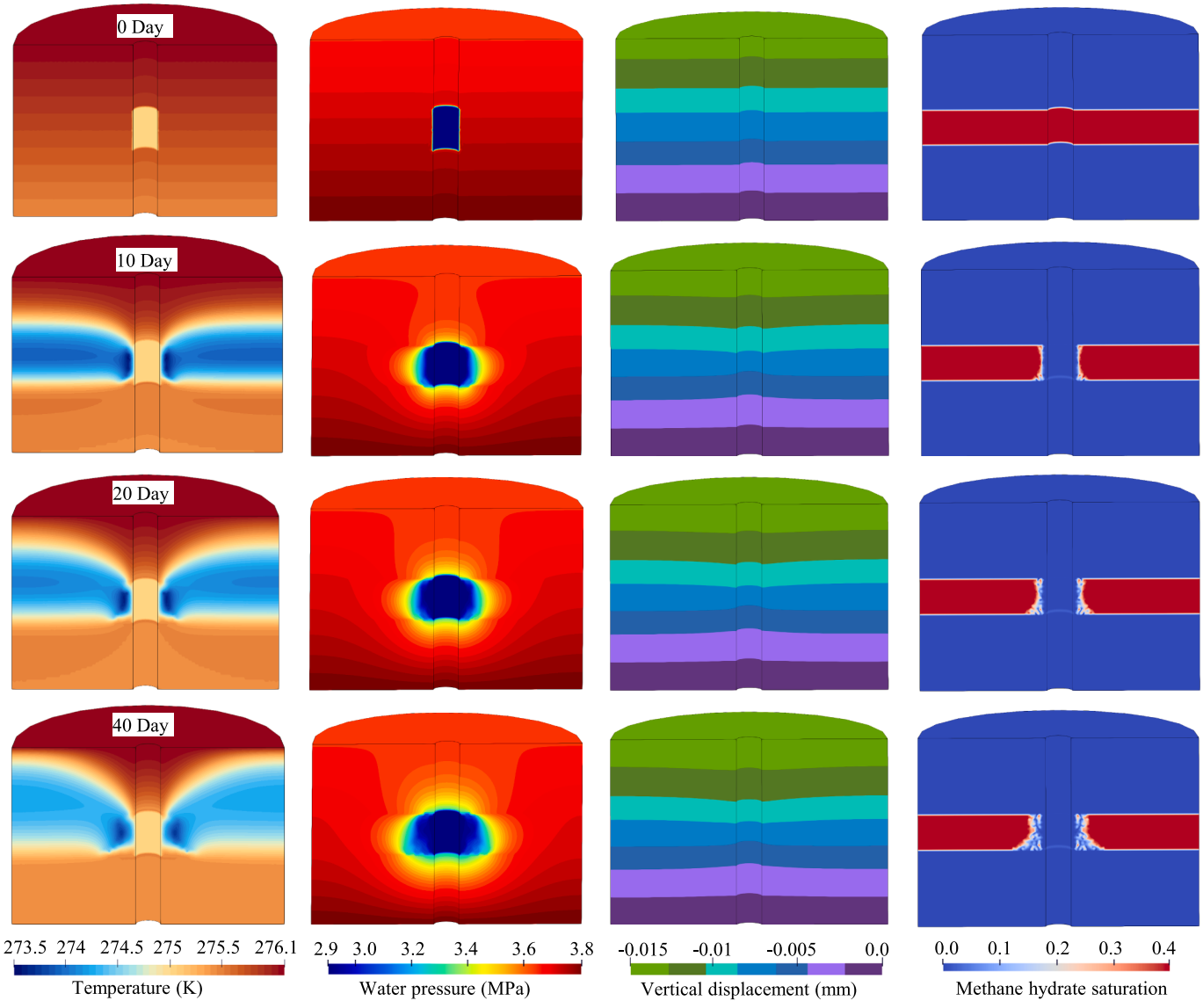

Fig. 17. Distributions of temperature, water pressure, vertical displacement and hydrate saturation at different time instants.

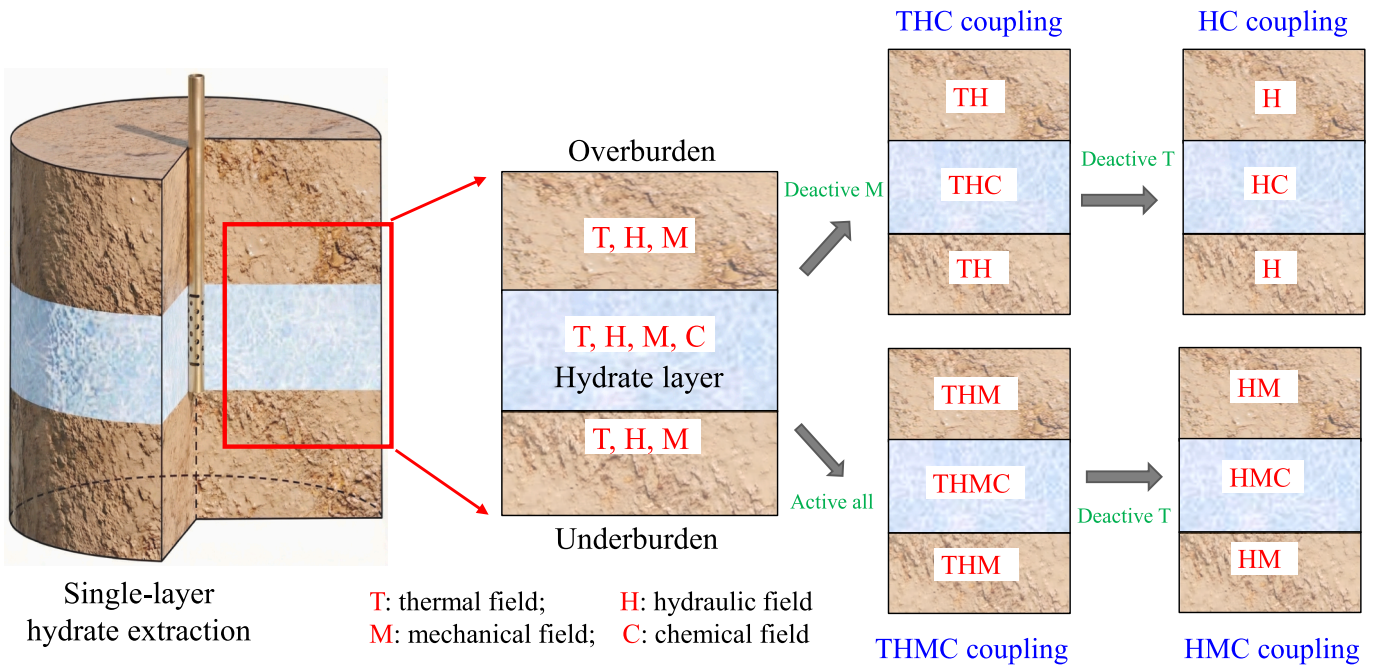

Fig. 18. A unified framework for the flexible evolution of various coupling schemes.

5.1. Comparative Analysis of Different Coupling Schemes

Fig. 18 illustrates the spatiotemporal flexibility of the proposed unified framework in tailoring multi-physics coupling schemes. Spatially, the framework accommodates the distinct physical characteristics of different model regions by assigning a THM coupling scheme to the overburden/underburden layers, while employing a fully coupled THMC scheme in the intermediate hydrate-bearing reservoir to account for the chemical (C) field of hydrate concentration. Temporally, the framework facilitates customized comparative analyses through the selective activation or deactivation of physical fields. By leveraging this capability to isolate certain fields, three simplified schemes (HC, HMC, and THC couplings) are conducted under strictly identical material properties and model configurations to elucidate the specific impacts of individual couplings on hydrate production behavior. The results were systematically compared to the fully THMC coupling case. Fig. 19 presents contour plots of hydrate saturation evolution for the four coupling schemes at selected times. The thermal field is found to exert the dominant control on the predicted response. Schemes excluding thermal effects (HC and HMC) exhibit extremely rapid dissociation, with high-saturation zones retreating swiftly until complete depletion. Notably, in the absence of considering the thermal effect, the depressurization front propagates unimpeded to regions distant from the production well, causing them to dissociate earlier and faster. This leads to a distinctive wedge-shaped hydrate saturation profile (resembling a “train-head”), characterized by extensive depletion in the upper section and lagging dissociation in the lower section. In contrast, schemes that account for thermal effects (THC and THMC) experience a significant slowdown in dissociation due to reservoir cooling caused by the endothermic nature of hydrate decomposition and the resulting “self-preservation” effect, leaving substantial undissociated hydrate. The mechanical field has only a minor influence on saturation morphology. Neglecting thermal effects thus severely overestimates production efficiency, highlighting the necessity of including thermal coupling (via THMC or THC models) for reliable predictions. Fig. 20 further highlights the differences in water pressure propagation across the four coupling schemes. At early production times (0.1hr), interactions between the different physical fields remain minimal, resulting in nearly identical pressure distributions across all schemes, governed primarily by hydraulic gradients. As production advances into intermediate and later stages (10-100hr), marked discrepancies emerge. The thermal effect imposes the strongest suppression on pressure propagation, resulting in substantially smaller low-pressure depletion zones for thermal-inclusive schemes (THC and THMC) compared to non-thermal ones. In particular, the THC and THMC schemes exhibit the most pronounced pressure retardation. This is attributed to the fact that reservoir cooling inhibits dissociation, thereby maintaining low permeability in the hydrate-bearing zones and restricting pressure transmission. These results conclusively show that omitting thermal coupling causes traditional models to overestimate pressure propagation rates and depressurization extents significantly, thereby inflating production forecasts.

Fig. 21(a) compares the cumulative gas production curves obtained from four different coupling schemes. The results reveal that schemes neglecting thermal effects (HC and HMC) yield significantly higher gas production volumes and rates than those incorporating thermal effects (THC and THMC). This finding confirms that the thermal field acts as a controlling factor in hydrate dissociation, whereas the mechanical field exerts negligible influence under the small-strain assumption. Fig. 21(b) presents the radial pressure distributions along the midline of the hydrate layer at selected times. Thermally coupled cases (dashed lines) consistently display higher pressures than isothermal cases (solid lines). This pressure elevation is attributed to localized temperature drop caused by endothermic dissociation; this cooling shifts the phase equilibrium, restricting further hydrate dissociation. Consequently, the presence of undissociated solid hydrate maintains low permeability, impeding fluid flow and preventing the formation of an effective pressure drawdown cone. In contrast, the isothermal conditions in the HC and HMC schemes promote sustained dissociation throughout the reservoir. During the early stage (within 30hr), rapid dissociation combined with low permeability impedes fluid drainage distal to the production well, leading to a temporary high-pressure plateau. As dissociation progresses and hydrate saturation decreases, the permeability increases, eventually dissipating this plateau in the later stages.

5.2. Effects of different exploitation schemes on hydrate production performance

In deep-sea environments, methane hydrates are often distributed

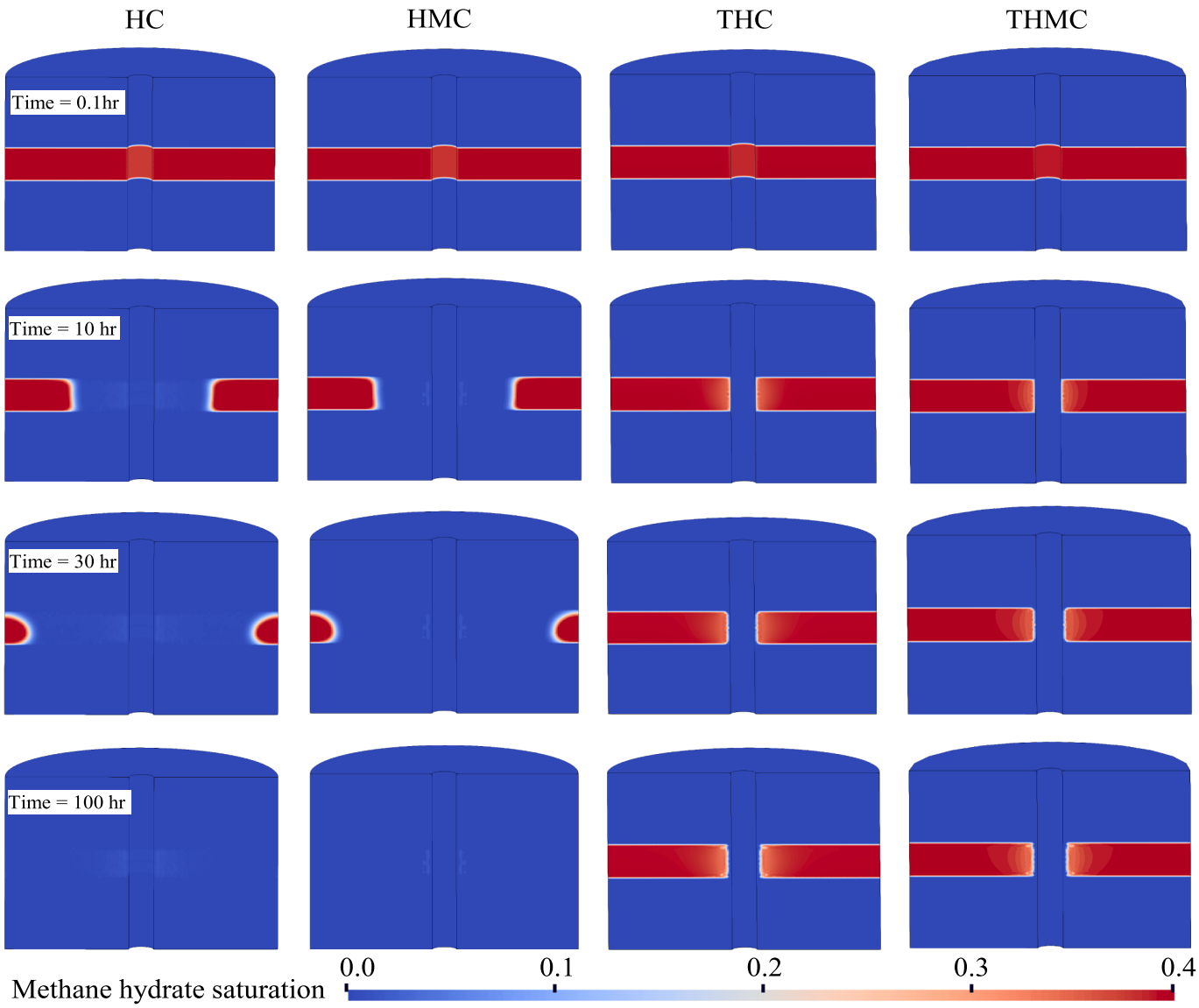

Fig. 19. Hydrate saturation distributions across coupling schemes at different time instants.

across multiple reservoirs at varying depths rather than being confined to a single layer (as illustrated in Fig. 22(a)). Consequently, investigating the interactions and multi-physics coupling mechanisms between vertically stratified hydrate-bearing layers during production is crucial for enhancing exploitation efficiency. Accordingly, this study explores the coupling processes in dual-layer hydrate reservoirs, using the model mesh and boundary conditions shown in Fig. 22(b). The mesh comprises 9306 elements and 4752 nodes. The overall model setup and material parameters remain consistent with the single-layer reservoir in Fig. 15 (c), except for the application of distinct pressure and temperature boundaries at the production outlets of the upper layer (Layer-A) and lower layer (Layer-B). To examine the effects of varying production pressure and temperature, five group tests were conducted (see Table 4), with the Base group serving as the reference. The remaining four groups all employed a combined depressurization-and-heating strategy, but with differentiated stimulation across the upper and lower hydrate-bearing layers. G-1 and G-2 applied both depressurization and heating simultaneously to only the upper or lower layer, respectively, to assess production response from single-layer stimulation. In contrast, G-3 and G-4 adopted cross-layer configurations: depressurization in Layer-A with heating in Layer-B (G-3), and the reverse (G-4). These configurations aim to elucidate the synergistic effects and compatibility of different driving mechanisms across layers, thereby enabling a systematic assessment of the optimal production strategy for dual-layer systems. In addition, point-A on the model's upper surface is monitored for vertical displacement to assess the engineering safety risks from production-related subsidence.

Fig. 23 illustrates the temporal evolution of pore water pressure in the dual-layer hydrate reservoir under different production strategies. Pressure perturbations propagate radially outward from the wellbore, initially forming a distinct depressurization zone near the well that expands over time. The Base group, limited by higher outlet pressure, exhibits the smallest pressure-affected region. In contrast, the depressurization strategies (G-1 to G-4) achieve significantly larger low-pressure zones. Notably, G-2 (depressurization in the lower Layer-B) demonstrates superior pressure transmission efficiency compared to G-1 (depressurization in the upper Layer-A). The optimal configuration is observed in G-4 (heating in Layer-A, depressurization in Layer-B), where intense lower-layer depressurization establishes a broad low-pressure zone at the base and effectively induces pressure reduction in the upper layer by overcoming interlayer resistance. This results in the widest depressurization extent and optimal interlayer connectivity, significantly outperforming G-3 (depressurization in Layer-A, heating in Layer-B). These findings underscore the critical role of lower-layer

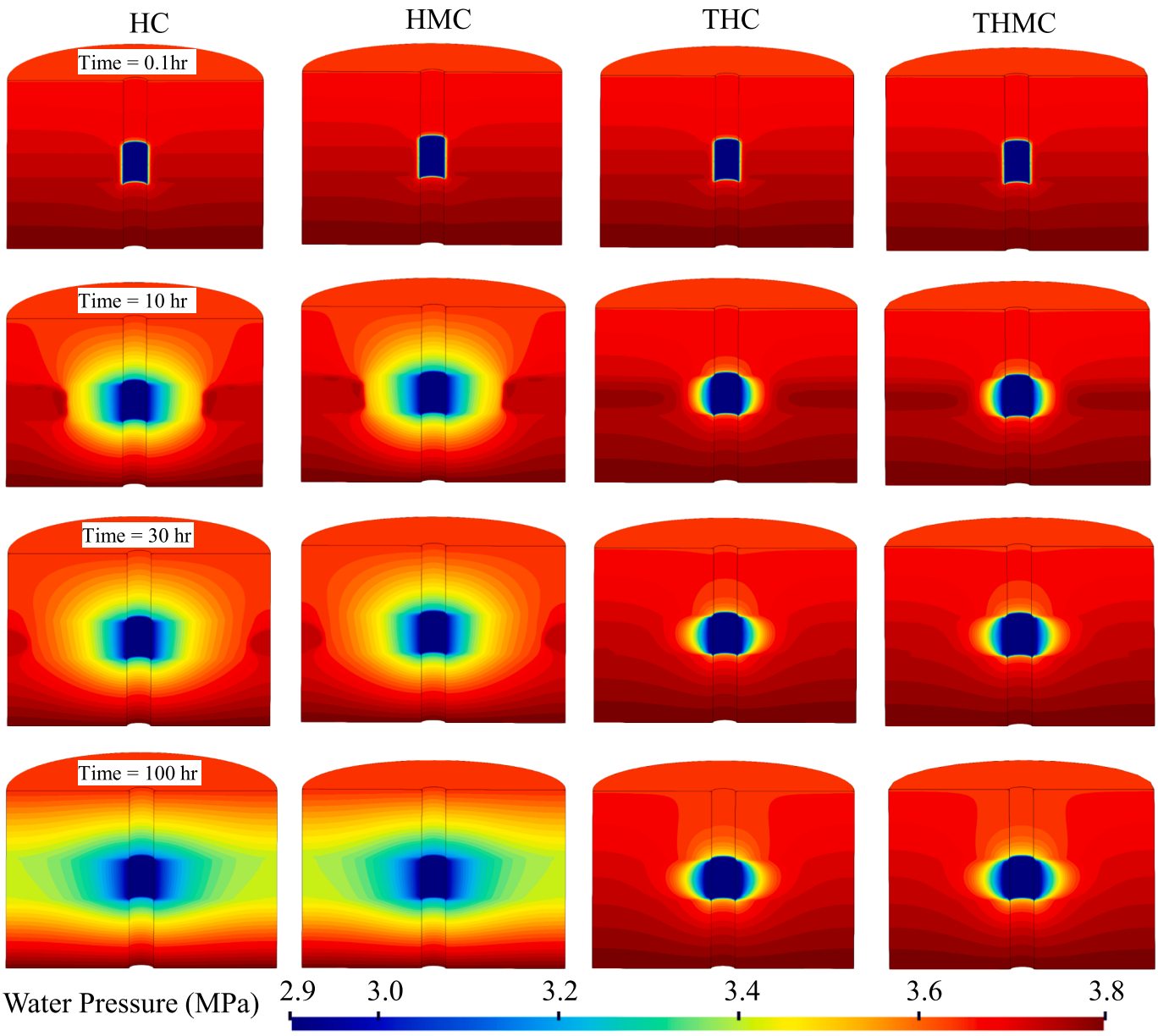

Fig. 20. Pore water pressure distributions across coupling schemes at different time instants.

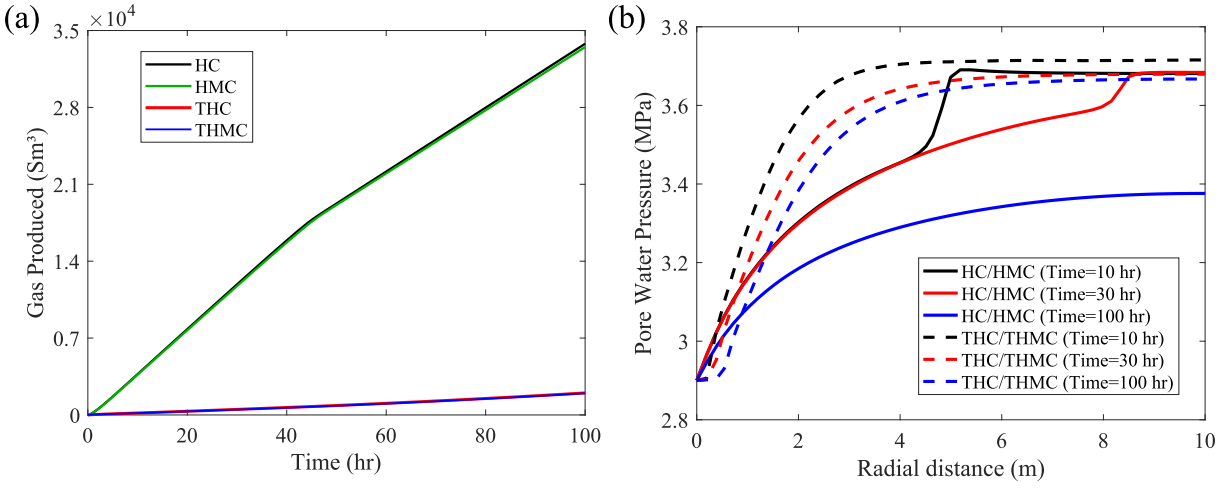

Fig. 21. (a) Gas production curves and (b) radial midline pressure distributions in the hydrate layer at different time instants under various coupling schemes.

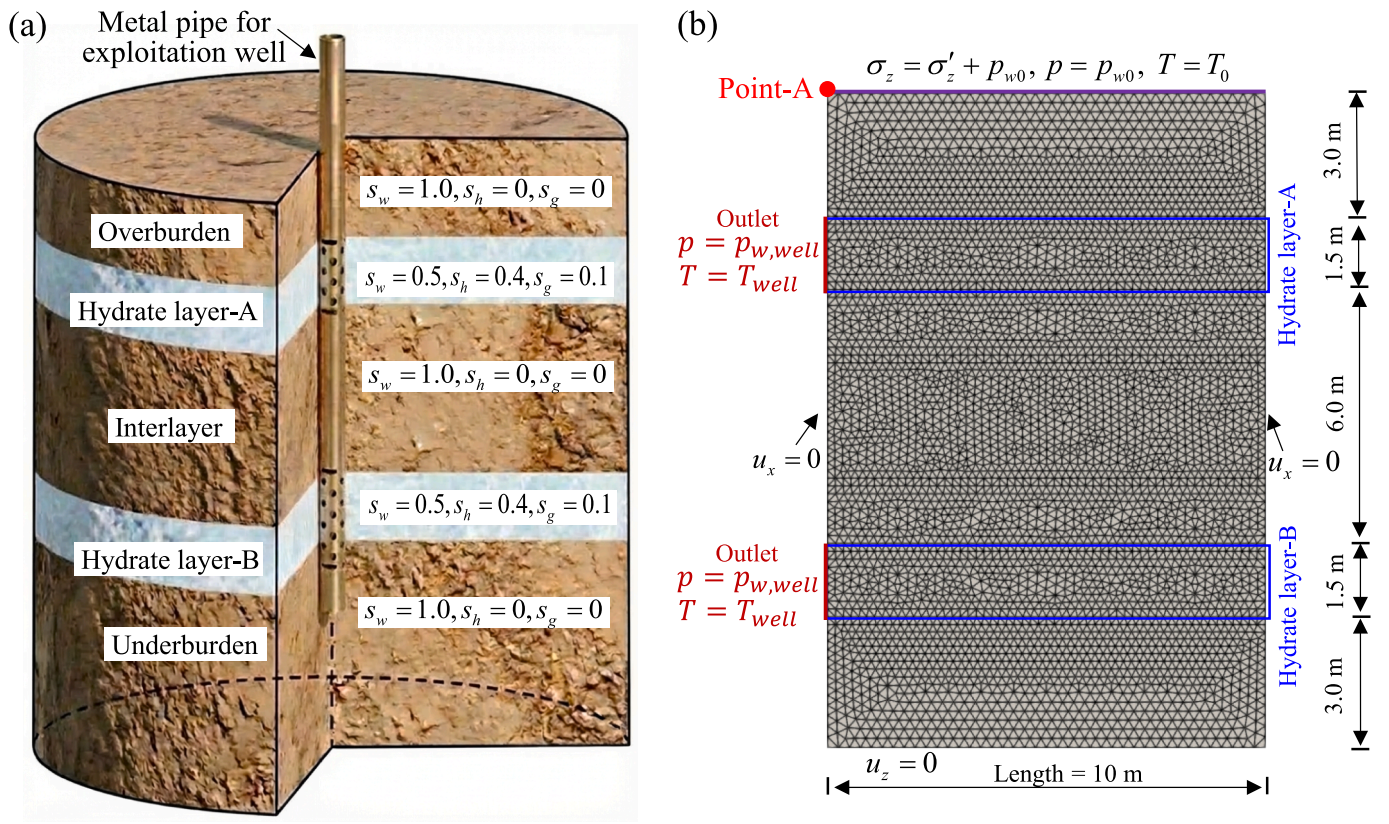

Fig. 22. (a) axisymmetric production model of dual-layer hydrate reservoirs and (b) discretized mesh and applied boundary condition.

Table 4
Pressure and temperature settings at the outlet of hydrate reservoirs.

Groups	Hydrate layer-A outlet pressure (MPa)	Hydrate layer-A outlet temperature (K)	Hydrate layer-B outlet pressure (MPa)	Hydrate layer-B outlet temperature (K)
Base	2.9	275	2.9	275
G-1	2.6	278	2.9	275
G-2	2.9	275	2.6	278
G-3	2.6	275	2.9	278
G-4	2.9	278	2.6	275

depressurization in driving efficient pressure propagation in dual-layer reservoirs.

Fig. 24 shows the temporal evolution of the temperature field in the dual-layer hydrate reservoir during production, governed by the competing effects of endothermic dissociation and applied thermal stimulation. Temperature perturbations exhibit a characteristic pattern of near-well sudden change followed by far-field propagation. During the early extraction phase (0.01 hr), both the heating region and the low-temperature region arising from endothermic decomposition remain localized near the well. As extraction progresses, the low-temperature funnel caused by endothermic dissociation expands significantly into the far field, while the high-temperature stimulated region remains localized around the well, serving as a continuous heat source. In the Base group, identical temperature and pressure boundary conditions are applied to both layers, resulting in similar cooling patterns. In contrast, the other groups show distinct low-temperature distributions due to differing boundary conditions between the upper and lower layers. At late stages (100 hr), the interlayer temperature distributions in G-2 and G-4 are notably higher than in G-1 and G-3, suggesting greater potential for continued dissociation. Furthermore, G-2 applies simultaneous depressurization and heating to the same layer, allowing sustained heat supply near the well to counteract cooling from dissociation and thereby accelerate hydrate dissociation. In contrast, G-4 (heating in the upper layer and depressurization in the lower layer) leaves the lower-layer low-temperature zone without rapid heat replenishment, resulting in slower dissociation. Accordingly, Fig. 25(a) shows that G-2 achieves the highest gas production, whereas G-4 exhibits the lowest enhancement efficiency relative to the Base group. Fig. 25(b) reveals that vertical displacement at point A follows a nonlinear settlement pattern, with rapid subsidence in the early stage followed by slower rates later. All enhanced production strategies significantly increase subsidence risk compared with the Base case. Among them, G-1 experiences the largest subsidence and highest risk, whereas G-2 has the smallest subsidence, indicating better stability while maintaining high recovery efficiency. Based on these results, it is recommended to exploit deeper reservoirs using an extraction strategy that combines lower pressures with higher temperatures, thereby optimizing recovery efficiency and formation stability.

The results presented above demonstrate that thermal effects not only influence hydrate dissociation behavior but also have important implications for field-scale production design. As shown in Fig. 25, endothermic cooling associated with hydrate dissociation can significantly affect the evolution of the reservoir response. Therefore, thermal coupling should be explicitly considered in production forecasting, since neglecting temperature variations may lead to an overestimation of gas recovery and dissociation efficiency. The simulations also indicate that

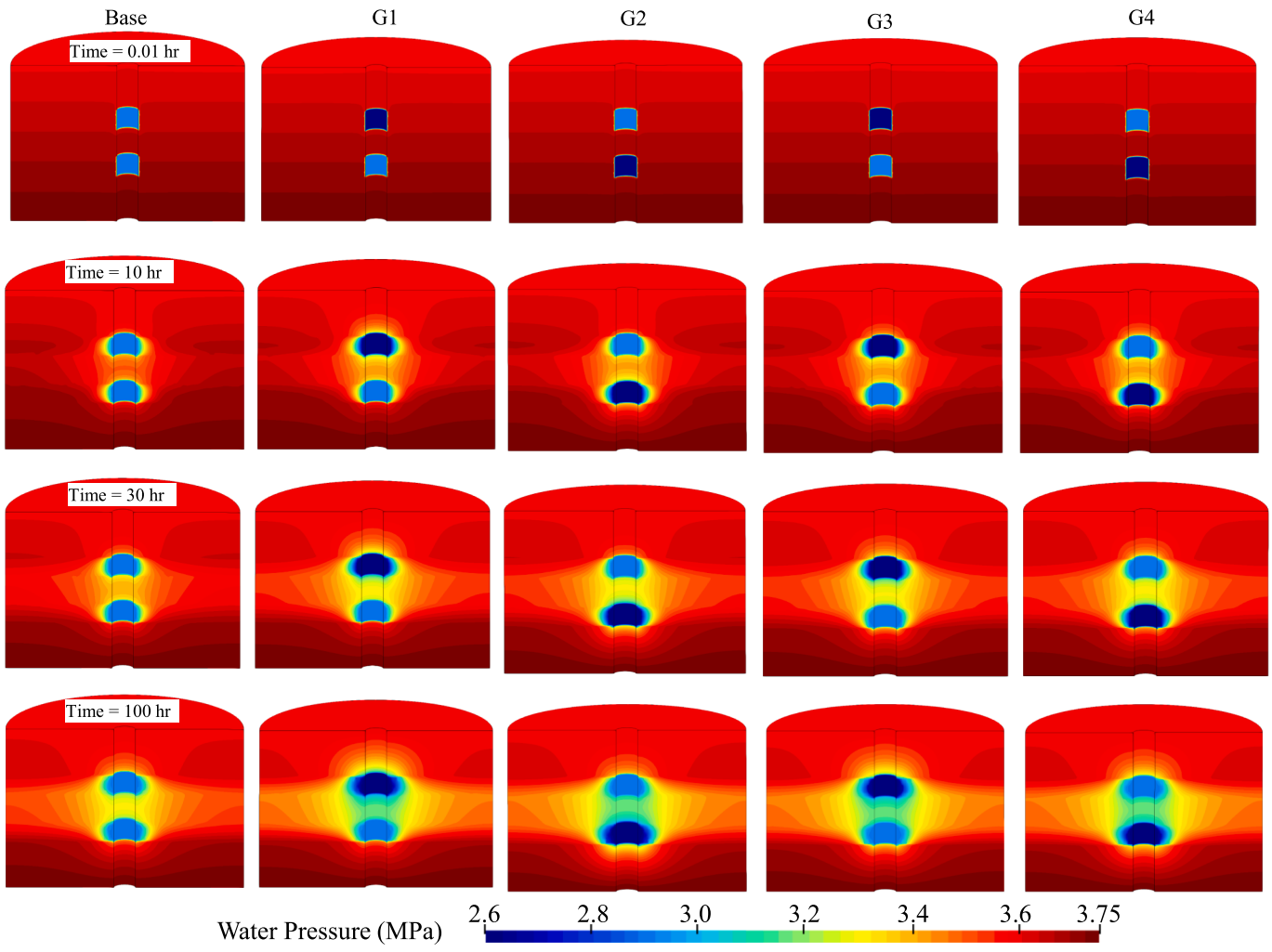

Fig. 23. Pore water pressure distributions across different groups at different time instants.

cooling effects can propagate over a relatively large region, whereas externally supplied heat remains more localized. This suggests that thermal stimulation is likely to be more effective when applied in a targeted manner near production wells or within selected hydrate-bearing layers rather than through uniform heating. In addition, production performance should be evaluated together with geomechanical responses. The coupled evolution of pore pressure, temperature, and vertical displacement provides important information for assessing reservoir stability and the potential risk of excessive consolidation or seafloor subsidence. For the dual layer reservoir, simultaneous depressurization and heating in the deeper hydrate layer provides a favorable strategy because it promotes continued dissociation while reducing the adverse influence of cooling and deformation on reservoir stability. Overall, these observations highlight the value of the proposed framework for supporting production strategy optimization, thermal stimulation design, and engineering risk assessment.

6. Conclusions

This study develops a unified computational framework to address the intricate multi-physics coupling and numerical instabilities in methane hydrate production. The defining spatiotemporal flexibility of this architecture enables seamless transitions between varying coupling degrees (e.g., THMC, THM, TH), allowing specific schemes to be locally assigned to distinct geological domains and dynamically adapted to evolving exploitation phases without reconstructing the overburden/underburden models. By incorporating bubble functions, this highly flexible framework effectively mitigates volumetric locking in low-permeability, near-undrained conditions, while offering superior accuracy and reduced sensitivity to mesh distortion. The integration of SUPG further extends the framework's applicability to convection-dominated problems by eliminating spurious oscillations.

The reliability of this unified framework was systematically verified through progressively complex benchmark tests. Initial two-phase flow and subsequent TM/THM simulations confirmed the integrated algorithms' capability to eliminate numerical oscillations and maintain high accuracy under challenging near-undrained conditions. Ultimately, validation against Masuda's experiment demonstrated the framework's comprehensive effectiveness in capturing the highly nonlinear multi-phase flow and phase transition behaviors inherent in fully coupled THMC processes. Ultimately, the current framework was applied to large-scale deep-sea methane hydrate production to investigate reservoir dynamics under various coupling scenarios and exploitation strategies. Results reveal that neglecting thermal effects significantly overestimates gas production because endothermic cooling inherently retards pressure propagation, while dissociation-induced skeletal weakening simultaneously exacerbates subsidence risks. To navigate these coupled thermos-mechanical constraints, comparative analyses of dual-layer reservoirs indicate that prioritizing simultaneous depressurization and heating in deeper strata provides the optimal engineering strategy for maximizing recovery and maintaining geomechanical stability.

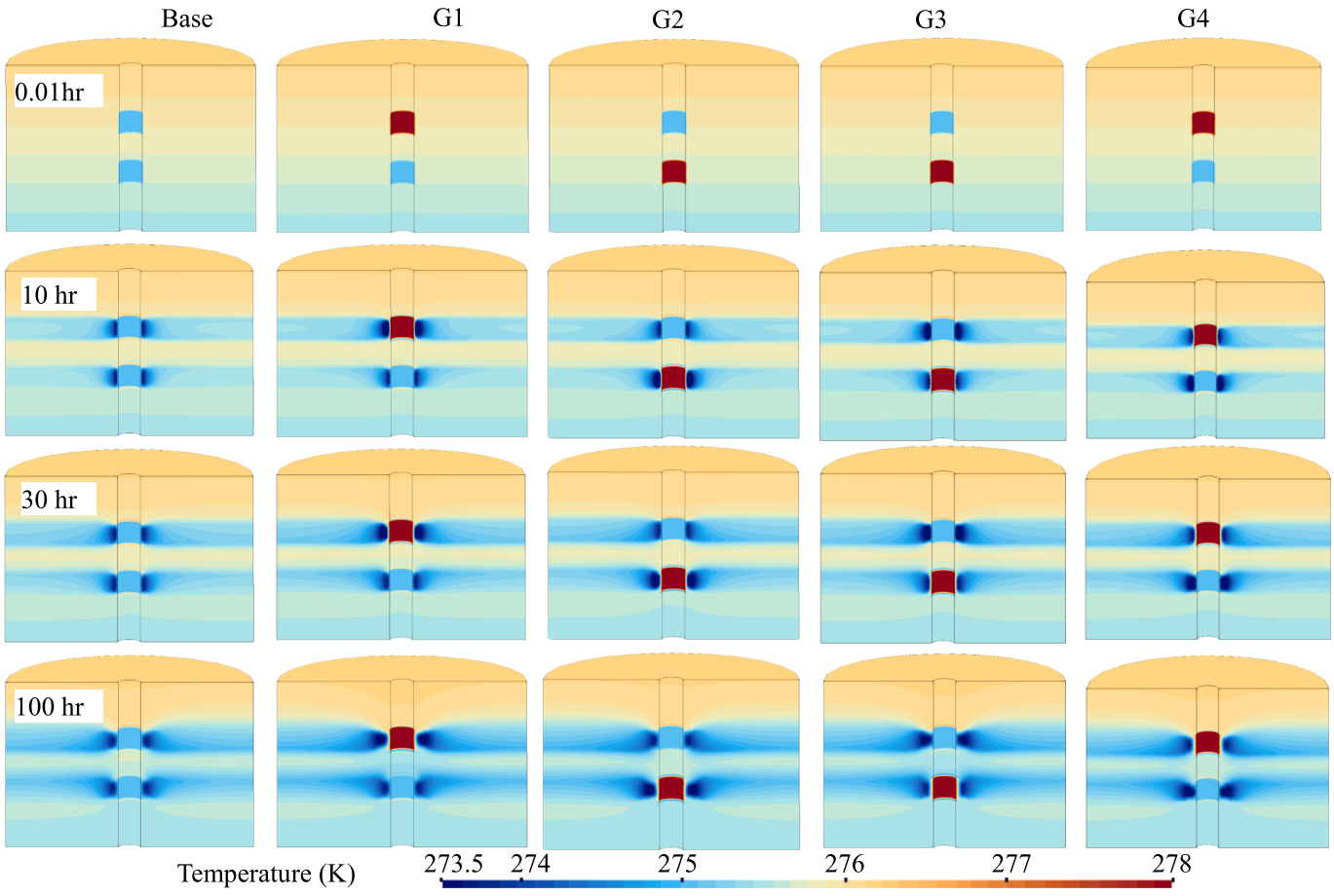

Fig. 24. Temperature distribution of different groups at different time instants.

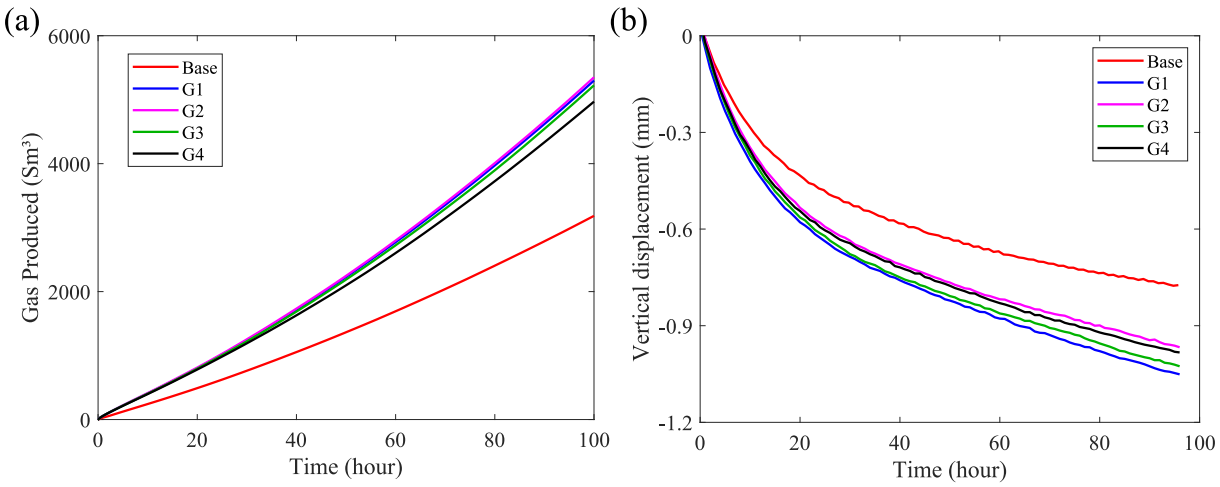

Fig. 25. (a) Gas production curves and (b) vertical displacement curve at point-A in the dual-layer hydrate system.

Future research will aim to improve the compatibility and applicability of the current computational framework by extending the present 2D method to a fully three-dimensional (3D) formulation (Zhang et al., 2025) and by incorporating advanced constitutive models (Cheng and Yin, 2025; Cheng and Yin, 2026) together with large-deformation theories (Qiu et al., 2026; He et al., 2025; Tang et al., 2025). Such an extension will inevitably increase the computational cost of coupled THMC simulations due to the substantial growth in degrees of freedom, memory requirements, and nonlinear solution complexity. While the current OpenMP based implementation is adequate for the present 2D analyses, large scale 3D applications will likely require more scalable strategies, such as distributed memory parallelization, domain decomposition, and efficient iterative solvers with robust preconditioners. Recent advances in geomechanics have further emphasized the importance of material heterogeneity, strain localization, and fracture evolution in governing the deformation and failure of porous geomaterials (Wang et al., n.d.; Liu et al., 2026; Liu and Wang, 2025). Accordingly, future developments should integrate these mechanisms into the

proposed multi-physics framework to enable more realistic simulations of hydrate-bearing sediments and other complex geomaterials under coupled THMC processes.

CRediT authorship contribution statement

Zi-Qi Tang: Writing – original draft, Visualization, Validation, Methodology. **Jie Yang:** Writing – review & editing, Supervision, Investigation, Formal analysis. **Qi Zhang:** Writing – review & editing, Validation, Methodology, Conceptualization. **Wang-Qi Xu:** Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The detailed derivation of the mass conservation equations governing each phase in porous media is presented as follows:

Solid grains,

$$\begin{aligned}
&\frac{\mathrm{d}((1-\phi)\rho_s)}{\mathrm{d}t}+(1-\phi)\rho_s\nabla\cdot\boldsymbol{v}_s=0\\
&\rho_s\frac{\mathrm{d}(1-\phi)}{\mathrm{d}t}+(1-\phi)\frac{\mathrm{d}\rho_s}{\mathrm{d}t}+(1-\phi)\rho_s\nabla\cdot\boldsymbol{v}_s=0\\
&\quad\Downarrow\mathrm{Eq.}(6):\frac{\mathrm{d}\rho_s}{\mathrm{d}t}=\rho_s\cdot(-3\alpha_s)\frac{\mathrm{d}T}{\mathrm{d}t}+\exp[-3\alpha_s(T-T_0)]\frac{\mathrm{d}\rho_0^s}{\mathrm{d}t}\approx\rho_s\cdot(-3\alpha_s)\frac{\mathrm{d}T}{\mathrm{d}t}\\
&\rho_s\frac{\mathrm{d}(1-\phi)}{\mathrm{d}t}-3\rho_s(1-\phi)\alpha_s\frac{\mathrm{d}T}{\mathrm{d}t}+(1-\phi)\rho_s\nabla\cdot\boldsymbol{v}_s=0\\
&\frac{\mathrm{d}\phi}{\mathrm{d}t}=(1-\phi)\nabla\cdot\boldsymbol{v}_s-3\alpha_s(1-\phi)\frac{\mathrm{d}T}{\mathrm{d}t},
\end{aligned}\tag{64}$$

Hydrate,

$$\begin{aligned}
&\frac{\mathrm{d}(\phi s_h\rho_h)}{\mathrm{d}t}+\phi s_h\rho_h\nabla\cdot\boldsymbol{v}_h=-M_hR_h\\
&\quad\Downarrow(\mathrm{Assuming}\,\rho_h=\mathrm{constant},\ \boldsymbol{v}_h=\boldsymbol{v}_s)\\
&\frac{\mathrm{d}s_h}{\mathrm{d}t}=-\frac{M_hR_h}{\rho_h\phi}-\frac{s_h}{\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t}-s_h\nabla\cdot\boldsymbol{v}_s\\
&\quad\Downarrow\frac{\mathrm{d}\phi}{\mathrm{d}t}=(1-\phi)\dot{\varepsilon}_v-3\alpha_s(1-\phi)\frac{\mathrm{d}T}{\mathrm{d}t}\\
&\frac{\mathrm{d}s_h}{\mathrm{d}t}=-\frac{M_hR_h}{\rho_h\phi}-\frac{s_h}{\phi}\dot{\varepsilon}_v+3\alpha_s(1-\phi)\frac{s_h}{\phi}\frac{\mathrm{d}T}{\mathrm{d}t},
\end{aligned}\tag{65}$$

Water,

$$\begin{aligned}
&\frac{\partial(\phi_w\rho_w)}{\partial t}+\nabla\cdot(\phi_w\rho_w\boldsymbol{v}_w)=6M_wR_h\\
&\quad\Downarrow\boldsymbol{q}_w=\phi s_w\widetilde{\boldsymbol{v}}_w=\phi_w(\boldsymbol{v}_w-\boldsymbol{v}_s)\\
&\frac{\partial(\phi_w\rho_w)}{\partial t}+\nabla\cdot(\phi_w\rho_w\boldsymbol{v}_s)+\nabla\cdot(\rho_w\boldsymbol{q}_w)=6M_wR_h\\
&\quad\Downarrow\mathrm{Material\ Derivative}:\frac{\mathrm{d}(\phi_w\rho_w)}{\mathrm{d}t}=\frac{\partial(\phi_w\rho_w)}{\partial t}+\boldsymbol{v}_s\cdot\nabla(\phi_w\rho_w)\\
&\frac{\mathrm{d}(\phi_w\rho_w)}{\mathrm{d}t}+\phi_w\rho_w\nabla\cdot\boldsymbol{v}_s+\nabla\cdot(\rho_w\boldsymbol{q}_w)=6M_wR_h\\
&\quad\Downarrow\mathrm{Assuming\ \ water\ \ is\ \ incompressible},\\
&\rho_ws_w\frac{\mathrm{d}\phi}{\mathrm{d}t}+\rho_w\phi\frac{\mathrm{d}s_w}{\mathrm{d}t}+\phi_w\rho_w\nabla\cdot\boldsymbol{v}_s+\rho_w\nabla\cdot\boldsymbol{q}_w=6M_wR_h,
\end{aligned}\tag{66}$$

Gas,

$$
\begin{aligned}
&\frac{\partial(\phi s_g\rho_g)}{\partial t}+\nabla\cdot(\phi s_g\rho_g\boldsymbol{v}_g)=M_gR_h\\
&\Downarrow \boldsymbol{q}_g=\phi s_g\widetilde{\boldsymbol{v}}_g=\phi s_g(\boldsymbol{v}_g-\boldsymbol{v}_s)\\
&\frac{\partial(\phi s_g\rho_g)}{\partial t}+\nabla\cdot(\phi s_g\rho_g\boldsymbol{v}_s)+\nabla\cdot\left(\rho_g\boldsymbol{q}_g\right)=M_gR_h\\
&\Downarrow\text{Material Derivative}:\ \frac{\mathrm{d}(\phi s_g\rho_g)}{\mathrm{d}t}=\frac{\partial(\phi s_g\rho_g)}{\partial t}+\boldsymbol{v}_s\cdot\nabla(\phi s_g\rho_g)\\
&\frac{\mathrm{d}(\phi s_g\rho_g)}{\mathrm{d}t}+\phi s_g\rho_g\nabla\cdot\boldsymbol{v}_s+\nabla\cdot\left(\rho_g\boldsymbol{q}_g\right)=M_gR_h\\
&\phi\frac{\mathrm{d}s_g}{\mathrm{d}t}+s_g\frac{\mathrm{d}\phi}{\mathrm{d}t}+\frac{\phi s_g}{\rho_g}\frac{\mathrm{d}\rho_g}{\mathrm{d}t}+\phi s_g\nabla\cdot\boldsymbol{v}_s+\nabla\cdot\boldsymbol{q}_g+\frac{\nabla\rho_g}{\rho_g}\boldsymbol{q}_g=\frac{M_gR_h}{\rho_g}\ ,
\end{aligned}
\tag{67}
$$

Gas-water mixture,

$$
\begin{aligned}
&\underbrace{\frac{\mathrm{d}s_g}{\mathrm{d}t}+\frac{s_g}{\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t}+\frac{s_g}{\rho_g}\frac{\mathrm{d}\rho_g}{\mathrm{d}t}+s_g\nabla\cdot\boldsymbol{v}_s+\frac{\nabla\cdot\boldsymbol{q}_g}{\phi}+\frac{\nabla\rho_g}{\phi\rho_g}\boldsymbol{q}_g-\frac{M_gR_h}{\phi\rho_g}}_{\text{gas mass conservation equation}}+\underbrace{\frac{\mathrm{d}s_w}{\mathrm{d}t}+\frac{s_w}{\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t}+s_w\nabla\cdot\boldsymbol{v}_s+\frac{\nabla\cdot\boldsymbol{q}_w}{\phi}-\frac{6M_wR_h}{\phi\rho_w}}_{\text{water mass conservation equation}}=0\\
&\qquad\qquad\Downarrow\text{using}\quad\frac{\mathrm{d}\rho_g}{\mathrm{d}t}=\frac{\rho_g}{p_g}\dot{p}_g-\frac{\rho_g}{T}\dot{T}\\
&\underbrace{(\dot{s}_g+\dot{s}_w)+(s_g+s_w)\nabla\cdot\boldsymbol{v}_s+\frac{(s_g+s_w)}{\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t}+\frac{s_g}{p_g}\dot{p}_g+\frac{\nabla\rho_g}{\phi\rho_g}\boldsymbol{q}_g+\frac{\left(\nabla\cdot\boldsymbol{q}_g+\nabla\cdot\boldsymbol{q}_w\right)}{\phi}-\frac{s_g}{T}\dot{T}-\frac{M_gR_h}{\phi\rho_g}-\frac{6M_wR_h}{\phi\rho_w}}_{\text{hybrid mass conservation equation for water - gas mixture}}=0\\
&\qquad\qquad\Downarrow\text{using}\quad\frac{\mathrm{d}\phi}{\mathrm{d}t}=(1-\phi)\nabla\cdot\boldsymbol{v}_s-3\alpha_s(1-\phi)\dot{T}\ \text{and}\ \nabla\cdot\boldsymbol{v}_s=\dot{\varepsilon}_v\\
&\frac{(s_g+s_w)}{\phi}\dot{\varepsilon}_v+\frac{s_g}{p_g}\dot{p}_g+\frac{\nabla\rho_g}{\phi\rho_g}\boldsymbol{q}_g+\frac{\left(\nabla\cdot\boldsymbol{q}_g+\nabla\cdot\boldsymbol{q}_w\right)}{\phi}-\frac{s_g}{T}\dot{T}-\frac{3\alpha_s(s_g+s_w)(1-\phi)}{\phi}\dot{T}-\dot{s}_h-\frac{M_gR_h}{\phi\rho_g}-\frac{6M_wR_h}{\phi\rho_w}=0\ .
\end{aligned}
\tag{68}
$$

Appendix B

Taking the classical T3 element as an example, the corresponding shape functions for each physical field are given by:

$$
\begin{aligned}
&\boldsymbol{N}_u=[\,\boldsymbol{N}_1\quad\boldsymbol{N}_2\quad\boldsymbol{N}_3\,]=\begin{bmatrix}N_1&0&N_2&0&N_3&0\\0&N_1&0&N_2&0&N_3\end{bmatrix},\\
&\boldsymbol{N}_p=\boldsymbol{N}_T=\boldsymbol{N}_\phi=[\,N_1\quad N_2\quad N_3\,],
\end{aligned}
\tag{69}
$$

and the gradients of the corresponding shape function are defined as follows:

$$
\begin{cases}
\boldsymbol{B}_u=[\,\boldsymbol{L}_u\boldsymbol{N}_1\quad\boldsymbol{L}_u\boldsymbol{N}_2\quad\boldsymbol{L}_u\boldsymbol{N}_3\,]\\
\boldsymbol{B}_p=[\,\boldsymbol{L}_\alpha\boldsymbol{N}_1\quad\boldsymbol{L}_\alpha\boldsymbol{N}_2\quad\boldsymbol{L}_\alpha\boldsymbol{N}_3\,]\\
\boldsymbol{B}_T=[\,\boldsymbol{L}_\alpha\boldsymbol{N}_1\quad\boldsymbol{L}_\alpha\boldsymbol{N}_2\quad\boldsymbol{L}_\alpha\boldsymbol{N}_3\,]\\
\boldsymbol{B}_\phi=[\,\boldsymbol{L}_\alpha\boldsymbol{N}_1\quad\boldsymbol{L}_\alpha\boldsymbol{N}_2\quad\boldsymbol{L}_\alpha\boldsymbol{N}_3\,]
\end{cases},\qquad
\left(\boldsymbol{L}_u=\begin{bmatrix}\frac{\partial}{\partial x}&0&\frac{\partial}{\partial y}\\0&\frac{\partial}{\partial y}&\frac{\partial}{\partial x}\end{bmatrix}^T,\ \boldsymbol{L}_\alpha=\begin{bmatrix}\frac{\partial}{\partial x}&\frac{\partial}{\partial y}\end{bmatrix}^T\right).
\tag{70}
$$

By substituting Eqs. (69)∼(70) into Eqs. (30), (32) and (34), the expressions for the terms in Eqs. (37)∼(39) are derived as follows:

$$\boldsymbol{K}_u=\int_\Omega\boldsymbol{B}_u^T[\boldsymbol{\sigma}]\mathrm{d}\Omega\ , \tag{71}$$

$$\boldsymbol{K}_{uw}=-\int_\Omega\boldsymbol{B}_u^T\boldsymbol{m}^T\boldsymbol{N}_p\mathrm{d}\Omega\ ,\quad\text{with}\ \ \boldsymbol{m}=[\,1\quad1\quad0\quad1\,]\ , \tag{72}$$

$$\boldsymbol{K}_{uT}=-\int_\Omega\alpha_s\boldsymbol{B}_u^T\boldsymbol{D}\boldsymbol{m}^T\boldsymbol{N}_T\mathrm{d}\Omega\ , \tag{73}$$

$$\boldsymbol{K}_{wu}=\int_\Omega\boldsymbol{N}_p^T\frac{(s_g+s_w)}{\phi}\boldsymbol{m}\boldsymbol{B}_u\mathrm{d}\Omega\ , \tag{74}$$

$$\boldsymbol{K}_{ww}=\int_\Omega\boldsymbol{N}_p^T\frac{s_g}{\rho_g}\frac{M_g}{RT}\boldsymbol{N}_p\mathrm{d}\Omega\ , \tag{75}$$

$$\boldsymbol{K}_w=\int_\Omega\left(\boldsymbol{N}_p^T\frac{\boldsymbol{q}_g^T}{\phi\rho_g}\frac{M_g}{RT}\boldsymbol{B}_p+\boldsymbol{B}_p^T\left(\frac{k_{rg}}{\phi\mu_g}+\frac{k_{rw}}{\phi\mu_w}\right)\boldsymbol{k}\boldsymbol{B}_p\right)\mathrm{d}\Omega\ , \tag{76}$$

$$\boldsymbol{K}_{wTT} = -\int_{\Omega} \boldsymbol{N}_p^T \left(\frac{s_g}{T} + \frac{3\alpha_s (s_g + s_w)(1-\phi)}{\phi} \right) \boldsymbol{N}_T \mathrm{d}\Omega \,, \tag{77}$$

$$\boldsymbol{K}_{wT} = -\int_{\Omega} \boldsymbol{N}_p^T \frac{\boldsymbol{q}_g^T}{\phi T} \boldsymbol{B}_T \mathrm{d}\Omega \,, \tag{78}$$

$$\boldsymbol{K}_{TT} = \int_{\Omega} \boldsymbol{N}_T^T \rho c \boldsymbol{N}_T \mathrm{d}\Omega \,, \tag{79}$$

$$\boldsymbol{K}_{T} = \int_{\Omega} k_T \boldsymbol{B}_T^T \boldsymbol{B}_T \mathrm{d}\Omega + \int_{\Omega} \boldsymbol{N}_T^T \left(\rho_g c_g \boldsymbol{q}_g^T + \rho_w c_w \boldsymbol{q}_w^T \right) \boldsymbol{B}_T \mathrm{d}\Omega \,, \tag{80}$$

$$\boldsymbol{F}^u = \int_{\Gamma} \boldsymbol{N}_u^T \bar{\boldsymbol{t}} \mathrm{d}\Gamma + \int_{\Omega} \boldsymbol{N}_u^T \boldsymbol{b} \, \mathrm{d}\Omega + \int_{\Omega} \frac{s_g}{s_w + s_g} \boldsymbol{B}_u^T \boldsymbol{m}^T \boldsymbol{N}_p \mathrm{d}\Omega \boldsymbol{P}_c \,, \tag{81}$$

$$\begin{aligned} \boldsymbol{F}^w = & -\int_{\Omega} \boldsymbol{B}_p^T \frac{k_{rg} \boldsymbol{k}}{\phi \mu_g} \boldsymbol{B}_p \mathrm{d}\Omega \boldsymbol{P}_c - \int_{\Omega} \boldsymbol{N}_p^T \frac{\boldsymbol{q}_g^T}{\phi \rho_g} \frac{M_g}{RT} \boldsymbol{B}_p \mathrm{d}\Omega \boldsymbol{P}_c \\ & - \int_{\Omega} \boldsymbol{N}_p^T \frac{s_g}{\rho_g} \frac{M_g}{RT} \dot{p}_c \mathrm{d}\Omega + \int_{\Omega} \boldsymbol{N}_p^T \left(\dot{s}_h + \frac{M_g R_h}{\phi \rho_g} + \frac{6 M_w R_h}{\phi \rho_w} \right) \mathrm{d}\Omega \\ & + \int_{\Omega} \boldsymbol{B}_p^T \left(\frac{\rho_g k_{rg} \boldsymbol{k}}{\phi \mu_g} + \frac{\rho_w k_{rw} \boldsymbol{k}}{\phi \mu_w} \right) \boldsymbol{g} \mathrm{d}\Omega - \int_{\Gamma} \boldsymbol{N}_p^T \frac{\bar{q}_g}{\phi} \, \mathrm{d}\Gamma - \int_{\Gamma} \boldsymbol{N}_p^T \frac{\bar{q}_w}{\phi} \, \mathrm{d}\Gamma \,, \end{aligned} \tag{82}$$

$$\boldsymbol{F}^h = -\int_{\Gamma_N} \boldsymbol{N}_T^T Q_N \mathrm{d}\Gamma + \int_{\Gamma_R} \boldsymbol{N}_T^T h_r (T_r - T) \mathrm{d}\Gamma - \int_{\Omega} \boldsymbol{N}_T^T M_h R_h \Delta H \mathrm{d}\Omega \,. \tag{83}$$

Appendix C

This appendix outlines the derivation of the simplified governing equations for the two-phase flow problem utilized to verify the efficacy of the SUPG method. To simplify calculations, the porosity is normalized to 1.0 and the compressibility of the two-phase fluids in the porous medium along with the effects of other phases and physical fields are neglected. Under these assumptions, the water mass conservation equation and the gas mass conservation equation are simplified to the following expressions:

$$\frac{\partial s_w}{\partial t} + \nabla \cdot \boldsymbol{q}_w = 0, \qquad \frac{\partial s_g}{\partial t} + \nabla \cdot \boldsymbol{q}_g = 0 \,. \tag{84}$$

Based on the Buckley-Leverett theory, the total flux in the porous medium is given by $\boldsymbol{q}_t = \boldsymbol{q}_w + \boldsymbol{q}_g$. It is assumed that the velocity of each phase is solely proportional to the square of its saturation with all other parameters (such as viscosity and intrinsic permeability) set to unity. Neglecting gravitational effects and assuming an isotropic medium, the velocities of the water and gas phases are further derived from Darcy's law as follows:

$$\begin{cases} \boldsymbol{q}_w = -s_w^2 \nabla p \\ \boldsymbol{q}_g = -s_g^2 \nabla p \end{cases} \Rightarrow \begin{cases} \boldsymbol{q}_w = s_w^2 \Big/ \left(s_w^2 + s_g^2 \right) \boldsymbol{q}_t \\ \boldsymbol{q}_g = s_g^2 \Big/ \left(s_w^2 + s_g^2 \right) \boldsymbol{q}_t \end{cases} . \tag{85}$$

Substituting Eq. (85) into the water mass conservation equation in Eq. (84) yields:

$$\begin{aligned} & \frac{\partial s_w}{\partial t} + \nabla \cdot \left(\frac{s_w^2}{s_w^2 + s_g^2} \boldsymbol{q}_t \right) = 0 \\ \Rightarrow \; & \frac{\partial s_w}{\partial t} + \frac{s_w^2}{s_w^2 + s_g^2} \nabla \cdot \boldsymbol{q}_t + \boldsymbol{q}_t \cdot \nabla \left(\frac{s_w^2}{s_w^2 + s_g^2} \right) = 0 \,. \end{aligned} \tag{86}$$

Fluid incompressibility implies $\nabla \cdot \boldsymbol{q}_t = 0$, so that Eq. (86) further simplifies as follows:

$$\frac{\partial s_w}{\partial t} + \boldsymbol{q}_t \cdot \nabla \left(\frac{s_w^2}{s_w^2 + s_g^2} \right) = 0 \,. \tag{87}$$

Data availability

Data will be made available on request.

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